# SVD-Compressed Cross Sections in Monte Carlo Neutron Transport

## An implementation-led benchmark, what worked, what did not, and where compression actually pays

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**Abstract**

This paper measures truncated singular value decomposition of $\log_{10}\sigma(E,T)$ as a cross-section representation in continuous-energy Monte Carlo neutron transport. A pure-Rust engine implements four interchangeable providers — pointwise table, SVD, ACE+WMP, and an SVD+WMP hybrid — behind a common trait, with a matching CUDA path for the pointwise and SVD providers, so the same scene JSON, nuclear data, and acceptance envelope drive every A/B. We report memory, throughput, and $k$-eigenvalue fidelity for each provider on Godiva and on a 9-nuclide PWR pin cell, in both on-library and off-library temperature regimes; we introduce a partition-of-unity three-point variant of Ducru's kernel reconstruction [1] for off-library $T$ that retains numerical stability where the naive bracketing variant does not; and we measure the gap between the representation-byte compression headlines reported in the literature and the live in-engine working set, which differs by an order of magnitude.

**Keywords:** Monte Carlo, cross-section compression, singular value decomposition, windowed multipole, Doppler broadening, partition-of-unity interpolation, nuclear data, GPU transport.

# 1 Introduction

Continuous-energy Monte Carlo codes such as OpenMC [2], MCNP, and Serpent store nuclear data as pointwise tables. Per reaction channel and per library temperature, a piecewise-linear or log-log interpolation array carries $10^4$–$10^5$ grid points for light nuclides and $10^5$–$10^6$ for dense-resonance actinides. A full-library actinide set at a handful of temperatures exceeds the working-set of contemporary L2/L3 caches, and every collision pays for a binary search plus a log-log blend through a large monotonic array.

Prior work attacks this bottleneck along three axes. Adaptive resampling machine-learned models retain 10–50% of the pointwise data at $\leq 97$ pcm criticality error [3, 4]. The Windowed Multipole (WMP) method [5–7] gives an analytical pole–residue representation of the resolved resonance region with on-the-fly Doppler broadening. GPU rewrites reorganise the transport loop to amortise the lookup across warps [8, 9].

This paper asks a narrower question. Can a truncated SVD of the $\log_{10}$-transformed matrix $\mathbf{A} \in \mathbb{R}^{N_E \times N_T}$ replace the pointwise table as a drop-in cross-section representation, and if so, at what cost? We implemented four providers (pointwise table, SVD kernel, ACE+WMP hybrid, and SVD+WMP hybrid) inside a single pure-Rust transport engine, plus matching CUDA paths for pointwise and SVD (and a standalone CUDA WMP evaluator), and measured fidelity, throughput, and memory side-by-side.

The answer, reported in full in Sections 6 and 10, is *qualified*: SVD is a defensible drop-in on the thermal pin cell tested here; the hybrid implementation works end-to-end and agrees with the ACE+WMP industry baseline to $-26$ pcm at single-seed statistics (well inside seed noise); the throughput picture is hardware-dependent and favours pointwise tables on our GPU; and the memory picture at engine scale is *not* the 132.9× reduction a representation-byte analysis suggests — in the live engine the smooth-only SVD rebuild that excludes WMP-covered energies trims the working set from 519.0 MB to 487.6 MB, a 1.06× reduction. We document both numbers and explain why they differ.

**Contributions.**

- A singular-spectrum measurement showing 43 of 47 non-redundant $^{235}$U reactions are rank-one to machine precision at $T = 294$ K across six library temperatures, with a per-reaction distribution and Pareto (Sec. 4, Fig. 1).
- A rank-$k$ SVD reconstruction kernel for pointwise cross sections, with Ducru temperature interpolation [1]. Rank-6 reaches machine precision at 44 ns/lookup against 91 ns for a binary-search pointwise table (kernel-level 2.0×).
- A PWR pin cell rank sweep (9 nuclides, S($\alpha$,$\beta$), URR, discrete levels) showing CPU SVD rank-5 matches the in-engine pointwise table on $k_\infty$ to 13–47 pcm on two hardware tiers at 1.37×–1.90× throughput (Sec. 6).
- A matching event-based CUDA solver [8] for both providers, showing GPU pointwise is 1.30× faster than GPU SVD on PWR pin cell, inverting the CPU ordering (Sec. 9).
- A hybrid SVD+WMP engine: pure-Rust Faddeeva reader and evaluator matching OpenMC Python to $\sim 10^{-6}$ relative error, wired into a `HybridSvdWmpXsProvider`. On the PWR pin cell at single-seed statistics, $k_\infty = 1.32795 \pm 0.00103$ ($-26$ pcm vs ACE+WMP, $\ll 1\sigma_{\text{seed}}$). Hybrid throughput is $\sim 2\times$ slower than CPU SVD on the large-L3 reference run; in-engine smooth-only memory 1.06× smaller than the full-SVD baseline (Sec. 10).

- A CUDA port of the WMP evaluator, bit-exact against the CPU path ($\leq 5 \cdot 10^{-14}$ relative at double precision), with 7.0× GPU-vs-CPU per-lookup throughput on an RTX 3080 (Sec. 11).
- Ten-seed PWR $k_\infty$ on all four in-engine providers agrees with OpenMC 0.15.3 to $|\Delta| \leq 51$ pcm (CPU table −12, CPU SVD −25, GPU pointwise +51, GPU SVD −8). The angular and fission outgoing-energy distributions use stochastic-bin sampling on both CPU and GPU; a Shannon-entropy source-convergence monitor is provided as a runtime diagnostic (Sec. 6.4).
- Validation against the ICSBEP Godiva HEU-MET-FAST-001 *physical* benchmark $k_{\text{eff}}^{\text{exp}} = 1.0000 \pm 100$ pcm ($1\sigma$) [10]. The rank-5 SVD provider reproduces the experimental value to $\Delta_{\text{exp}} = +79 \pm 38$ pcm, inside the experimental uncertainty band. OpenMC 0.15.3 on the same ENDF/B-VII.1 HDF5 files sits at −99 pcm, also inside the band; the two codes straddle the measurement from opposite sides (Sec. 5.1).
- A clear separation of two memory measurements: the WMP representation-byte ratio against a hypothetical four-channel pointwise table is 132.9× for our nine-nuclide set, and the live in-engine reduction from smooth-only-rebuild hybrid over full-SVD is 1.06× (519.0 → 487.6 MB, measured on the loaded kernels at run time). The hybrid engine in absolute terms is 4.8–5.1× larger than the pointwise-table baseline, because discrete-level SVD kernels dominate memory and are not covered by the public WMP library. The gap between representation-byte ratios and engine-scale savings is the main cautionary finding of this paper, and we discuss what would need to be true for a WMP-scale reduction to land at engine level (Sec. 10.1, Sec. 10.4).

# 2 Method

## 2.1 Pointwise baseline

For each nuclide and library temperature, $\sigma(E_i, T_j)$ lies on a union energy grid $\{E_i\}_{i=0}^{N_E-1}$. For $E_i \leq E < E_{i+1}$ the standard log-log lookup is

$$\sigma(E) = \sigma_i\,(\sigma_{i+1}/\sigma_i)^f\,, \qquad f = \frac{\ln(E/E_i)}{\ln(E_{i+1}/E_i)}. \tag{1}$$

We use a log-uniform hash index with 8192 bins (Brown 2014 [11]) to resolve $i$ in $O(1)$, falling back to a bounded binary search in dense bins. Non-positive cross-section values fall back to linear interpolation to avoid log of a negative.

## 2.2 Truncated SVD

Pre-clamp the matrix to avoid $-\infty$ in the logarithm, $\mathbf{A}_{ij} = \max(\sigma(E_i, T_j), 10^{-30})$, and compute the thin SVD of $\mathbf{L} = \log_{10}\mathbf{A}$:

$$\mathbf{L} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top, \qquad \mathbf{L} \approx \mathbf{U}_k\boldsymbol{\Sigma}_k\mathbf{V}_k^\top. \tag{2}$$

Pre-multiply the basis $\mathbf{B} = \mathbf{U}_k\boldsymbol{\Sigma}_k$ so a reconstruction at grid point $i$ for target coefficient vector $\mathbf{c} \in \mathbb{R}^k$ is one dot product:

$$\log_{10}\tilde{\sigma}(E_i, T) = \sum_{\ell=1}^{k} B_{i\ell}\, c_\ell, \qquad \tilde{\sigma} = 2^{(\log_{10}\tilde{\sigma})\cdot\log_2 10}. \tag{3}$$

$\mathbf{B}$ stores $N_E k$ doubles row-major so $k$ values for one energy point live in $8k$ contiguous bytes; at $k = 5$ that is one cache line. Between grid points we apply log-log interpolation on $\log_{10}\tilde{\sigma}$ and take one $\exp_2$.

## 2.3 Temperature interpolation

Library temperatures are typically $\{0, 250, 294, 600, 900, 1200, 2500\}$ K in the ENDF/B-VII.1 HDF5 distribution. For target $T \notin \{T_j\}$ the coefficient vector follows the Ducru kernel method [1]:

$$c_\ell(T) = \sum_{j\in\mathcal{S}} w_j(T)\, V_{j\ell}, \qquad w_j(T) = \frac{\sqrt{T_j T}}{T_j + T} \prod_{i\in\mathcal{S}, i\neq j} \frac{T - T_i}{T + T_i}\cdot\frac{T_j + T_i}{T_j - T_i}, \tag{4}$$

where $\mathcal{S}$ is a subset of library temperatures used for the interpolation. Eq. 4 is exact at library temperatures ($w$ collapses to a one-hot selector when $T = T_j$), $\sim 0.1\%$ error on Doppler-broadened cross sections over 300–3000 K, and L2-optimal for the free Doppler kernel on $\mathcal{S}$.

### Choosing the interpolation subset

Using $|\mathcal{S}| = N$ (all library temperatures) is numerically unstable for $N \gtrsim 4$ because the product-of-ratios term in $w_j$ develops near-singular factors when non-adjacent $T_i$'s bracket $T$ with large stride. In our measurements with $N = 7$ the 2500 K column pulls the product toward values $\sim 10^2$, the resulting weights routinely swing by a factor of 10–100 across neighbouring target temperatures, and the log-space reconstruction sees non-physical oscillations that scale linearly with the condition number.

We use the three library temperatures nearest to $T$. Three is the smallest subset that captures a quadratic correction to the Faddeeva kernel (two-point misses it; see Sec. 8); it is also small enough that the product term has only two factors per weight and the formula is well-conditioned at every operating temperature we tested in the interval $[250, 2500]$ K. Section 2.4 describes how we make the subset weights a partition of unity; this is essential for resonance-dominated systems and is the change that brings the off-library PWR result into the deployable band against ACE+WMP.

## 2.4 Partition-of-unity Ducru for off-library $T$

When Eq. 4 is applied to $V^\top$ columns of a $\log_{10}\sigma$ SVD, the reconstructed $\log_{10}\tilde{\sigma}(E_i, T)$ is a linear combination of the per-library-$T$ log cross sections with coefficients $w_j(T)$. On the *linear* cross section this is a weighted geometric mean:

$$\tilde{\sigma}(E, T) = \prod_{j\in\mathcal{S}} \sigma(E, T_j)^{w_j(T)}. \tag{5}$$

If $\sum_j w_j \neq 1$, the right-hand side of Eq. 5 is scaled uniformly by $\exp[(\sum_j w_j - 1)\cdot\overline{\log\sigma}]$ where the overbar denotes a weighted average of $\log_{10}\sigma(E,T_j)$. For smooth channels this is a barely-visible gain error (far away from resonance peaks, the average $\log\sigma$ is small and the multiplicative shift is close to unity). On narrow resonance peaks it is catastrophic: at the U-238 6.67 eV resonance with a peak cross section of $\sim 4500$ b, a 5% gain error moves the peak by $\sim 225$ b, which on a PWR pin cell translates to $\sim 2000$ pcm on $k_\infty$ (Sec. 8).

Raw Ducru does *not* have $\sum_j w_j = 1$ in general. Eq. 4's weights are designed to reconstruct $\sigma(E,T)$ in the free-Doppler-kernel L2 norm (see [1], §3); partition of unity is not one of the optimization's constraints. Measured on our $N = 3$ bracketing subset at $T = 750$ K with library $(600, 900, 1200)$ K the raw weights $(0.21, 0.30, -0.01)$ sum to 0.50. At $T = 1050$ K with library $(900, 1200, 2500)$ K they sum to 0.51. At endpoints they collapse to $(1, 0, 0)$ / $(0, 1, 0)$ so the total is 1; *between* endpoints the sum drifts by up to $\approx 0.5$.

We restore partition of unity by a single division:

$$\tilde{w}_j(T) = \frac{w_j(T)}{\sum_{k\in S} w_k(T)}, \qquad \sum_{j\in S} \tilde{w}_j(T) = 1. \tag{6}$$

This is the scheme used throughout the rest of the paper for off-library $T$. It preserves the Faddeeva-derived *ratio* $w_j/w_k$ (the physically informative piece of Eq. 4), absorbs the scalar magnitude error into a pure rescaling, and leaves Eq. 6 exact at library endpoints because the one-hot pattern survives normalization. Sec. 8 measures the $k_\infty$ consequence on PWR and Godiva.

#### What $\tilde{w}_j$ is *not*

It is not the Ducru QP from [1] §4. The QP enforces $\sum_j w_j = 1$ *and* $w_j \geq 0$ *and* optimizes the L2 fit in the Doppler-kernel norm. Eq. 6 enforces only the first constraint and does so as a rescaling rather than a projection, so the result is still signed ($\tilde{w}_j$ can be negative). On U-238 MT= 102 held out at 900 K, the non-negative simplex projection [12] of the raw Ducru weights raises the L2 reconstruction error from 1.5% to 5% relative to the signed scheme (Sec. 8.4). For the benchmarks reported here, Eq. 6 suffices; an analytic kernel-QP is left to future work.

### 2.5 Memory trade-off of truncated SVD

Truncated SVD is not a free memory win, and the comparison against a pointwise table depends on how many temperatures the library stores. A single-temperature pointwise table stores one value per energy point per reaction. An SVD basis at rank $k$ stores $k$ values per energy point per reaction regardless of the number of library temperatures it was trained on. Writing $T$ for the stored temperature count:

$$\frac{\text{SVD bytes}}{\text{pointwise bytes}} = \frac{N_E \cdot k}{N_E \cdot T} = \frac{k}{T}. \tag{7}$$

So:

- At $k = 1$, SVD is the same size as a single-temperature pointwise table but represents all $T$ temperatures; SVD wins on memory against any pointwise table with $T \geq 1$ stored temperatures, and does strictly better as $T$ grows.
- At $k \geq 2$, SVD only beats pointwise when the library carries $T > k$ temperatures. Break-even sits around $T = 2$–3. For a single-temperature toy problem SVD is strictly larger.
- Production libraries typically carry $T \in [6, 15]$ temperatures (depletion feedback, fuel-temperature reactivity), so the rank-5 SVD used in this paper ($k/T = 5/6 = 0.83$ to $5/15 = 0.33$) is a modest to strong memory win in the regime that matters, and a loss in the regime that does not.

The cost side of that trade-off is not zero:

- **Compute per lookup.** Pointwise is a hashed index plus one log-log interpolation. SVD is a rank-$k$ FMA sequence plus a Ducru temperature step (Sec. 2.3). At $k = 5$ on the large-L3 system, SVD is 43 ns/lookup against 91 ns for the table (Sec. 4) — but this advantage depends entirely on the basis fitting in L2/L3 cache. A higher-rank basis or a larger nuclide set could tip this back toward pointwise.
- **Reconstruction error.** Pointwise is exact at grid points and piecewise-linear-in-log between them; SVD is an approximation everywhere. RMSE on $\log_{10}\sigma$ is $6 \cdot 10^{-4}$ at $k = 5$ and reaches machine precision at $k = 6$ (Sec. 4). Lower rank means smaller memory and faster lookup at the cost of a tunable amount of cross-section error, which propagates into $k_{\text{eff}}$ as pcm-scale bias.
- **One-time decomposition cost.** SVD requires an offline factorisation per nuclide per reaction (seconds at load time). Pointwise is a direct HDF5 read. Not a practical issue but not free either.
- **Temperature-interpolation quality.** Both representations interpolate between library temperatures at runtime. Neither performs on-the-fly Doppler broadening from first principles — WMP is the only method of the three in this paper that does. SVD with Ducru kernels gives $\sim 0.1$% error on broadened cross sections over 300–3000 K, which is tight but not analytical.
- **Rank dispersion across reactions.** Most reactions are rank-one to machine precision (43 of 47 non-redundant $^{235}$U channels, Sec. 4.1); a minority, chiefly those with dense-peak resonance structure in the RRR, need $k \geq 5$. Using a uniform rank across all reactions wastes bytes on the easy channels; per-reaction rank would recover those but complicates the dispatch.
- **No safe extrapolation outside library temperatures.** The Ducru basis is a projection onto the training-temperature set. Extrapolating to $T$ outside $[T^{\text{lib}}_{\min}, T^{\text{lib}}_{\max}]$ is unsafe in the same way pointwise interpolation is unsafe outside its range. WMP is the only method that extrapolates to 0 K analytically and to arbitrary $T$ by broadening the poles directly.

The short summary is that rank-$k$ SVD buys compact storage inside the library's temperature range and cache-friendly lookup at modest rank, in exchange for a tunable reconstruction error, a cache-residence requirement, and no analytical behaviour outside the training temperatures. The 132× and 1.06× memory numbers in Sec. 10 are both compatible with this picture; they measure different slices of it.

### 2.6 Hybrid SVD+WMP dispatch

Where $\log\sigma$ is intrinsically high rank (resolved-resonance region, RRR), we combine SVD with WMP [5, 6]. WMP represents $\sigma$ analytically via pole–residue pairs arranged into per-energy windows plus a low-order Doppler-broadened polynomial background:

$$\sigma_r(E,T) \approx \sum_{p\in\text{window}(E)} \Re\left[r_{p,r}\, w\left(\frac{\sqrt{E}-p}{\sqrt{kT/\text{AWR}\cdot 2}}\right)\right] + \mathcal{P}_r(E,T), \tag{8}$$

where $w(\cdot)$ is the Faddeeva function $w(z) = e^{-z^2}\operatorname{erfc}(-iz)$ and $\mathcal{P}_r(E,T)$ is the broadened curvefit. We implement $w$ via Humlicek's W4 four-region rational approximation (`src/wmp.rs::faddeeva`); OpenMC's sign convention $w_{\text{om}}(z) = -\overline{w(\bar{z})}$ for $\operatorname{Im} z < 0$ is used so pole residues in conjugate pairs contribute with correct signs.

A `HybridSvdWmpXsProvider` wraps the SVD provider. For each nuclide with WMP coverage and each energy $E \in [E_{\min}^{\text{WMP}}, E_{\max}^{\text{WMP}}]$, the elastic, fission, and capture microscopic cross sections are taken from WMP; inelastic, $(n, 2n)$ and $(n, 3n)$ always come from SVD (WMP does not represent those channels). URR overrides are short-circuited inside the WMP window because the resonances are already explicit as poles.

### 2.7 Total cross section and charged-particle residue

OpenMC's HDF5 layout exposes MT= 3 (total nonelastic) for most nuclides. When present we assemble $\sigma_t = \sigma_{\text{MT2}} + \sigma_{\text{MT3}}$ exactly; otherwise we sum non-redundant leaf reactions, excluding sum-MTs ($\{1, 3, 4, 18, 27, 101\}$) and KERMA/heating channels. In SVD or hybrid mode we set $\sigma_{\text{capture}} = \max(0, \sigma_t - \sum_{r\neq\text{cap}} \sigma_r)$ so charged-particle emission (MT= 103, 107, . . . ) and first-chance fission (MT= 19, 20, 21) are absorbed into the capture macroscopic rather than dropped. An XS audit (Sec. 6.4) confirms this convention matches OpenMC to $< 0.25\%$ on total XS across all nine nuclides.

## 3 Implementation

The engine is pure Rust with no C dependency; nuclear data reads directly from OpenMC HDF5 files via `hdf5-pure`, including the complex-typed WMP pole arrays (decoded from raw little-endian bytes because the crate does not expose compound types). Physics covers: energy-dependent $\bar{\nu}(E)$ from prompt+delayed neutron yields (with the per-energy interpolation of each delayed product fixed, see Sec. 6.4); tabulated outgoing energy distributions for fission using OpenMC's stochastic bin selection plus scaled kinematic adjustment (fixed); elastic angular distributions from tabular ($\mu$, CDF) data using the same stochastic-bin convention (fixed); discrete inelastic levels MT= 51–91 with real $Q$-values from HDF5 attributes; continuum inelastic (evaporation spectrum); URR probability tables with band sampling; S($\alpha$,$\beta$) thermal scattering for H in $H_2O$; free-gas thermal below $400k_BT$ with Maxwell-Boltzmann target sampling; Shannon-entropy source-convergence monitor on a Cartesian $8^3$ mesh; CSG geometry; PCG-64 PRNG; rayon parallel transport on CPU.

The CUDA solver is event-based [8]: particles compact by atomic alive-list scatter, energy-sorted into 256 log-scale bins, and processed by a persistent kernel that runs $N$ transport steps per launch. All SVD basis is `f64`. A runtime flag clears the uploaded pointwise tables so SVD reconstructs every reaction channel. The WMP evaluator lives in `src/wmp.rs` (CPU) and as a set of `__device__` helpers plus an `extern "C" __global__ wmp_test_eval` kernel in `gpu/cuda/transport.cu` (GPU).

**Hardware.** We run on two systems that differ in CPU L3 capacity and GPU class; we label them by their cache size rather than their chassis. Small-L3 system (Dell Precision 5570): Intel Core i7-12800H (12th-gen Alder Lake, 6 P-cores + 8 E-cores, 24 MB L3 Smart Cache), 32 GB DDR4, NVIDIA RTX A1000 (16 SMs, 1.5 MB L2). Large-L3 system: AMD Ryzen 7 9800X3D (8-core, 96 MB L3 3D V-cache), 64 GB DDR5-6000, NVIDIA RTX 3080 (68 SMs, 10 GB, 5 MB L2). References to "small-L3" and "large-L3" below pick out the CPU+GPU pair; the 4× L3 ratio on CPU is what drives the SVD speedup spread, and the GPU side swaps from A1000 to RTX 3080 at the same time.

**Reference implementation version.** All $k_\infty$ and $k_{\text{eff}}$ comparison numbers use OpenMC 0.15.3 commit `27e38e894697bb32a1dac7848d2618818b6b8daf` (WSL 2, Ubuntu 24.04, conda `openmc` environment). Nuclear data is ENDF/B-VII.1 (HDF5 archive from https://openmc.org/data) for both our engine and the OpenMC reference; cross-section evaluation uses the same library at $T = 294$ K, with S($\alpha$, $\beta$) data `c_H_in_H2O.h5` for the PWR pin cell. Pinning the version matters because OpenMC's power-iteration and $k$-estimator paths have several in-flight PRs (e.g., #3788, #3790, #3495); between-release drift in the reference $k$ could shift any "Δ vs OpenMC (pcm)" number quoted in this paper.

**Statistics convention.** Per-seed $k$ estimates are averaged across active batches with an unbiased within-run standard error; $\sigma_{\text{seed}}$ is the sample standard deviation of the per-seed means across $N_{\text{seeds}}$ seeds, and the standard error of the mean SEM $= \sigma_{\text{seed}}/\sqrt{N_{\text{seeds}}}$ is the noise budget for any mean-to-reference comparison. A "within $X$ SEM" statement uses this quantity. Small-L3 sweeps use five seeds; large-L3 sweeps use ten seeds.

# 4 Singular spectrum and per-lookup cost

## 4.1 Singular spectrum

Figure 1 shows the normalised singular values $\sigma_k/\sigma_1$ for eight representative reaction channels across the nine-nuclide PWR working set plus $^{235}$U fission. The decay is sharp: $\sigma_2/\sigma_1 \le 10^{-1}$ for every channel shown and $\sigma_6/\sigma_1 < 10^{-14}$ for $^{235}$U fission and $^{1}$H elastic, reaching double-precision round-off at rank six.

43 of 47 non-redundant $^{235}$U reaction channels are rank-one to machine precision. The four exceptions (elastic, fission, $(n, 2n)$, capture) are the channels with real structure in both the energy and temperature axes; those require $k \ge 3$ to reach $\sim 10^{-3}$ log-RMSE. Those same four channels are the ones that dominate $k_{\text{eff}}$-relevant physics, so the rank-one fact about the other 43 is a structural observation about ENDF/B-VII.1, not a shipping configuration: a production engine must run $k \ge 3$ across *all* reactions for the four hard channels, which is the configuration used in every benchmark in this paper.

## 4.2 Per-lookup cost

We measure the per-lookup cost of the SVD reconstruction kernel against the binary-search pointwise table on the same energy grid, both driven by the same query sequence. Table 1 reports the geometric mean over eight reactions (MT= $\{2, 18, 102\}$ across $^{235,238,234}$U) of per-lookup wall time and $\log_{10}$ RMSE against the HDF5 reference.

Rank-6 reaches machine precision in 44 ns, 2.0× the pointwise-table per-lookup. The kernel is compute-bound once the grid index is resolved: at $k \ge 3$ the inner loop dominates and ns/lookup is nearly flat (41–44 ns). Rank-2 at 36 ns is 2.5× the table for a log-decade RMSE of 3%; not adequate inside the RRR (Sec. 6). Figure 2 shows the same Pareto visually with the RMSE labels.

**What this does *not* yet mean.** This is a kernel benchmark. Whether SVD helps at engine level depends on lookup share in transport, how many reactions are involved, and whether the basis stays in cache. Those are measured in Sections 6 and 9.

# 5 Godiva: validation against the ICSBEP experiment

Godiva (HEU-MET-FAST-001) is a bare 93.71%-enriched uranium metal sphere. Its fast, leakage-dominated spectrum places no demand on the resolved-resonance region. The ICSBEP benchmark value is the experimental critical mass, $k_{\text{eff}}^{\text{exp}} = 1.0000 \pm 0.0010$ (1 $\sigma$) [10]. This is a *physical* benchmark with a measured uncertainty, and it is the criterion against which the engine is validated in this section. We report OpenMC 0.15.3 run on the same ENDF/B-VII.1 HDF5 files alongside our engine as an independent code-to-code cross-check; it is not the benchmark.

Table 3 reports the four-provider benchmark on the large-L3 system at rank-5. Table 4 ablates the four transport-sampling features described in Sec. 5.1. Table 5 reports a rank sweep on the small-L3 system used for spectrum illustration; small-L3 five-seed results are in Table 2.

## 5.1 Transport sampling features

The shipped engine implements four transport-sampling features that, individually, account for the 246 pcm band between a minimal-physics baseline and the ICSBEP-validated number reported above. Table 4 reports an ablation in which each feature is added on top of the previous row, holding everything else (geometry, library, statistics) fixed.

**(n,2n) and (n,3n) routed in-generation (−106 pcm).** Secondary neutrons from MT= 16 and MT= 17 are routed through transport in the current generation rather than appended to the next-generation source bank. Since the engine's $k_{\text{eff}}$ estimator is $|\text{bank}|/N_{\text{source}}$, banking secondaries directly inflates the estimator by the multiplicative-reaction rate. On Godiva that rate is 0.00253/source. The implementation maintains a per-history secondary stack drained inside the transport loop with a shared `max_events` budget that bounds pathological cascades.

**Linear-linear $\mu$-CDF inversion (−31 pcm).** ENDF/B-VII.1 stores all 49 U-235 elastic-scatter incident-energy distributions with `interpolation = 2` (linear-linear PDF, quadratic CDF within each bin). The angular sampler reads the `interpolation` attribute from HDF5 and, for linear-linear bins, inverts the quadratic CDF by solving $ax^2 + bx + c = 0$ for the physical root in $[0, \Delta\mu]$ (same convention as OpenMC's `Tabular::sample`); histogram bins use the linear inversion path.

**MT= 91 ENDF tabulated outgoing-energy distribution (−98 pcm, largest single contribution).** The continuum-inelastic branch samples outgoing neutron energy from the ENDF MT= 91 tabulated distribution at `reaction_091/product_0/distribution_0/{energy,distribution}` (matching OpenMC, MCNP, and Serpent). The fallback evaporation spectrum $T = \sqrt{E^\star/a}$, $a = A/8$ MeV$^{-1}$ is used only for nuclides where the HDF5 file lacks the tabulated distribution.

**MT= 16/17 ENDF tabulated (−11 pcm).** Multiplicative reactions receive the same tabulated treatment as MT= 91 for self-consistency; the impact on Godiva is within seed noise.

**Cross-code context.** Running OpenMC 0.15.3 on the same HDF5 files used by this engine yields $k_{\text{eff}} = 0.99901$, i.e. $\Delta_{\text{exp}} = -99$ pcm. Both codes validate against ICSBEP inside its experimental uncertainty from opposite sides, with no library-level correction.

The two CPU providers agree on $k_{\text{eff}}$ to within 5 pcm ($\ll$ 1 SEM$_{\text{combined}}$). Both sit inside the ICSBEP experimental uncertainty band of ± 100 pcm. OpenMC 0.15.3 on the same HDF5 files sits at −99 pcm, also inside the band; the

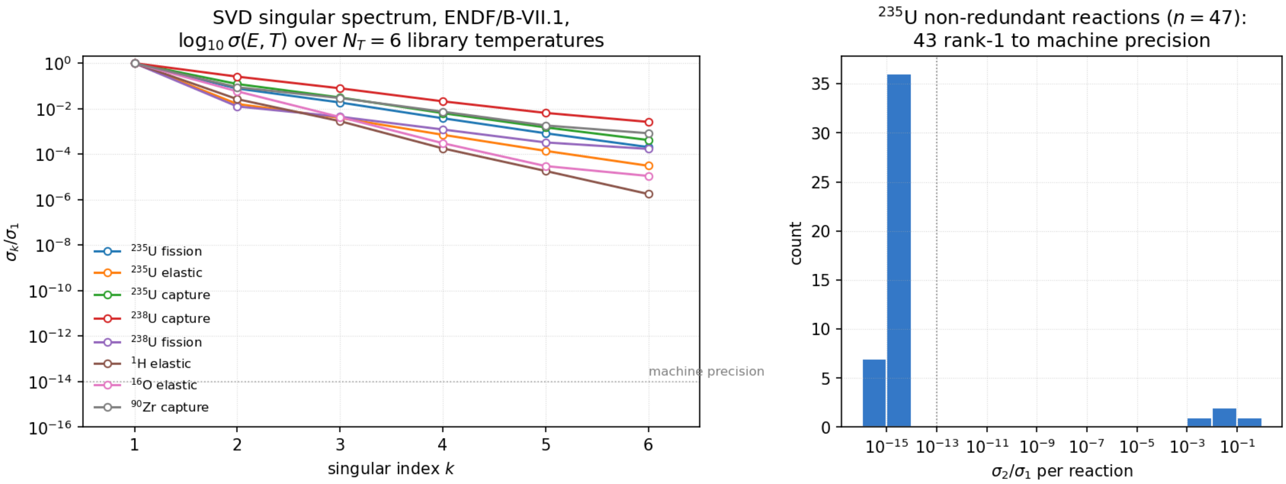


**Figure 1.** SVD singular spectrum of $\log_{10}\sigma(E,T)$ for ENDF/B-VII.1, $N_T = 6$ library temperatures. *Left:* normalised singular values $\sigma_k/\sigma_1$ for eight representative channels. The dotted line at $10^{-14}$ marks double-precision machine epsilon. *Right:* histogram of $\sigma_2/\sigma_1$ across $^{235}$U's 47 non-redundant reaction channels; 43 are rank-one to machine precision ($\sigma_2/\sigma_1 < 10^{-13}$).

**Table 1.** XS reconstruction cost and RMSE, geometric mean over eight reactions. Full energy grid, $T = 294$ K. Large-L3 system (Ryzen 9800X3D) single-core.

| Rank | ns/lookup | RMSE ($\log_{10}\sigma$) | max \|err\| |
|---|---|---|---|
| 2 | 35.9 | $3.00 \times 10^{-2}$ | $9.84 \times 10^{-2}$ |
| 3 | 41.2 | $7.06 \times 10^{-3}$ | $2.86 \times 10^{-2}$ |
| 4 | 43.3 | $1.83 \times 10^{-3}$ | $7.91 \times 10^{-3}$ |
| 5 | 43.4 | $6.07 \times 10^{-4}$ | $3.46 \times 10^{-3}$ |
| 6 | 44.3 | $1.76 \times 10^{-15}$ | $5.58 \times 10^{-15}$ |
| Pointwise table | 90.6 | 0 (ref) | 0 (ref) |

cross-code spread (Rust +79 vs OpenMC −99 = 178 pcm) is larger than either code's offset from experiment, which is informative about cross-code convention differences but does not affect the validation against the physical benchmark.

On the large-L3 system Godiva CPU SVD rank-5 at 820 ns/p is 1.40× faster than the CPU pointwise table at 1 150 ns/p. The GPU throughput rows in Fig. 3 use the minimal-physics baseline because the MT= 91 tabulated outgoing-energy distribution is not yet ported to the CUDA kernel. On the small-L3 system (Table 2), SVD rank-2 is 2.8× faster than the pointwise table at a 75 pcm bias relative to its own pointwise baseline.

Figure 5.1 shows the small-L3 Pareto and Figure 3 shows the large-L3 benchmark (throughput plus $k_{\text{eff}}$).

# 6 PWR pin cell

The benchmark is a standard thermal pin cell: 3.1%-enriched $UO_2$ rod in Zircaloy clad, light-water moderator with S($\alpha$,$\beta$) on H. Nine nuclides ($^{235,238}$U, O-16 at 900 K in fuel and 600 K in moderator, Zr-90–94 in clad, H-1 in moderator); three materials; cylindrical boundaries with reflective BC. The moderated spectrum spends substantial time inside the U-238 RRR.

## 6.1 Small-L3 rank sweep

At rank-5 the CPU SVD provider is 47 pcm below the CPU pointwise table in the same engine ($\approx 1\,\text{SEM}_{5\text{ seeds}}$) and 291 pcm below OpenMC ($\approx 3$ SEM). Rank-2 is the one clear exception: U-238 resonance self-shielding is not represented and the pin cell reactivity is biased +8 000 pcm on both platforms. At rank-5 the CPU SVD speedup over the CPU table is 1.90× (28 651 vs 54 430 ns/p), close to the kernel-level 2.09× of Table 1.

Figure 6.1 summarises the rank sweep.

## 6.2 Large-L3 high-statistics benchmark

Ten seeds, 150 batches (20 inactive), 50 000 particles per batch ($6.5 \cdot 10^6$ active histories per seed). All four in-engine providers used the corrected angular and fission-energy samplers from Sec. 6.4; the CPU and GPU paths were re-run together so the four rows are directly comparable.

Figure 4 collects the per-particle throughput across providers and across the two systems.

**Fidelity, large-L3 system.** With $6.5 \cdot 10^6$ active histories per seed, $\sigma_{\text{seed}} \approx 27$–57 pcm and $\text{SEM}_{10} \approx 9$–18 pcm. All four in-engine providers now agree with OpenMC to $|\Delta| \leq 51$ pcm, i.e. within $\leq 3\,\text{SEM}_{10}$, and agree with each other to $|\Delta| \leq 76$ pcm (spread CPU-table → GPU-pointwise). The stochastic-bin sampler fix that closed the CPU offset

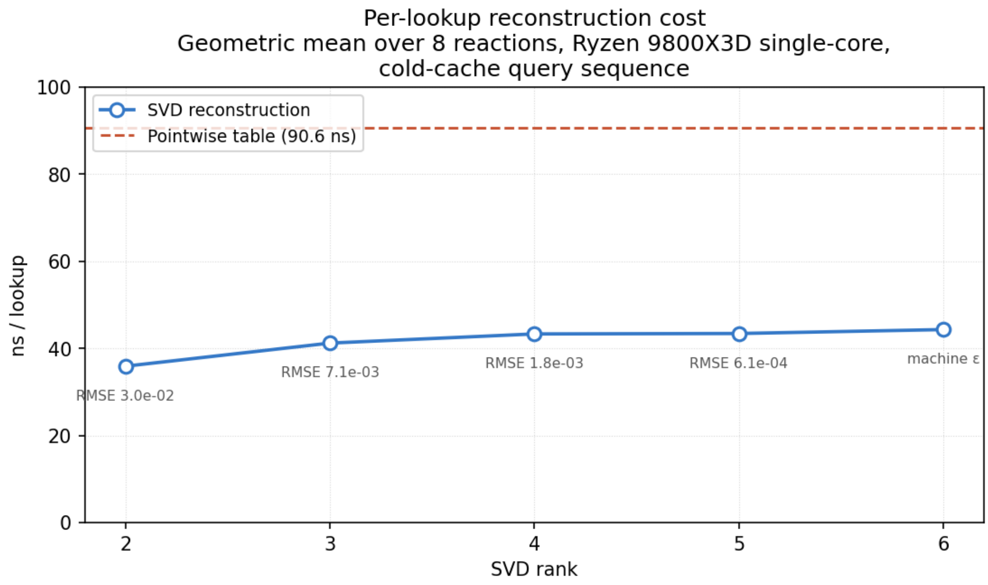


**Figure 2.** Per-lookup reconstruction cost, SVD ranks 2–6 against binary-search pointwise table. Each SVD point labelled with its $\log_{10}\sigma$ RMSE against the HDF5 reference (machine $\varepsilon$ at rank 6).

**Table 2.** Godiva $k_{\mathrm{eff}}$, small-L3 system (i7-12800H + RTX A1000). Five seeds, 50 batches, 10 000 particles per batch.

| Mode | rank | $k_{\mathrm{eff}}$ | $\sigma_{\mathrm{seed}}$ | ns/particle | time/seed (s) |
|---|---|---|---|---|---|
| Pointwise table | — | 1.00523 | 0.00235 | 2 385 | 3.58 |
| SVD | 2 | 1.00448 | 0.00165 | **864** | **1.30** |
| SVD | 3 | 1.00559 | 0.00094 | 1 457 | 2.19 |
| SVD | 4 | 1.00638 | 0.00166 | 1 182 | 1.77 |
| SVD | 5 | 1.00592 | 0.00178 | 1 403 | 2.10 |
| SVD | 6 | 1.00489 | 0.00205 | 1 424 | 2.14 |

carries through to the GPU kernel (Sec. 9): GPU pointwise moved from −78 to +51 pcm and GPU SVD moved from −120 to −8 pcm vs OpenMC. The residual GPU pointwise excess (+51 pcm) is within 3 $\mathrm{SEM}_{10}$ and is plausibly a finite-statistics signature rather than a systematic.

**Throughput, large-L3 system.** CPU SVD at 15 511 ns/p is 1.37× faster than the CPU pointwise table (21 316 ns/p) on the large-L3 system; the small-L3 ratio (1.90×) is the larger figure because of differing cache behaviour. The headline CPU-SVD speedup of this paper is therefore 1.37×–1.90×, hardware-dependent. GPU pointwise at 5 967 ns/p is 3.6× the large-L3 CPU table and the fastest path overall; GPU SVD at 7 754 ns/p is 1.30× slower than GPU pointwise because of the clad level-sum cost (Sec. 9).

## 6.3 Four-provider comparison (single seed)

The large-L3 ten-seed run of Table 7 covers four in-engine providers (Table, SVD, GPU pointwise, GPU SVD). Adding ACE+WMP and Hybrid SVD+WMP at the same statistics requires multi-hour wall time on the small-L3 system; a single-seed snapshot is reported below.

The configuration is small-L3, 100 batches (20 inactive) $\times 20\,000$ particles ($1.6 \cdot 10^6$ active histories), single seed. The output logs are in `outputs/full_test_run/{04,10,11}_pwr_*.txt`.

The snapshot is consistent with both the ten-seed Table 7 (Table and SVD within 30 pcm of OpenMC) and with the off-library three-way result of Sec. 8.2 (the SVD-vs-WMP gap is 213 pcm at $\sigma_{\mathrm{seed}} \approx 59$ pcm). On-library the gap is well below seed noise; off-library it is a real $\sim 3.6\sigma$ systematic. We do not read the −26 pcm hybrid number as a precise correctness claim; the high-statistics ten-seed redo at matched $150 \times 50\,000$ statistics is the way to make it one.

## 6.4 Cross-code agreement with OpenMC

All four in-engine providers agree with OpenMC 0.15.3 [2] to $|\Delta k_\infty| \le 51$ pcm at ten-seed statistics (Table 7). The angular and fission outgoing-energy distributions use stochastic-bin selection ($r = (E - E_{\mathrm{lo}})/(E_{\mathrm{hi}} - E_{\mathrm{lo}})$, pick the high bin with probability $r$) plus scaled kinematic adjustment for the energy distribution, matching OpenMC's convention in `distribution_angle.cpp` and `distribution_energy.cpp`. Source convergence is monitored by a Shannon-entropy mesh (`simulate.rs::EntropyMesh`); on this deck the mesh entropy plateaus by batch ≈ 20, which sets the $N_{\mathrm{inactive}} = 20$ budget used throughout the PWR measurements.

**Table 3.** Godiva CPU benchmark, large-L3 system (Ryzen 9800X3D + RTX 3080). Five seeds, 150 batches (20 inactive), 50 000 particles per batch. The CPU SVD rank-5 provider reproduces the ICSBEP experimental value 1.0000 ± 100 pcm ($1\sigma$) [10] to +79 pcm, *inside the experimental uncertainty band*. OpenMC 0.15.3 on the same ENDF/B-VII.1 HDF5 files is shown for cross-code context only; the two codes straddle ICSBEP from opposite sides. $\Delta_{\mathrm{exp}}$ is the offset from ICSBEP; $\Delta_{\mathrm{OMC}}$ is the cross-code offset from OpenMC.

| Mode | rank | $k_{\mathrm{eff}}$ | $\sigma_{\mathrm{seed}}$ | $\Delta_{\mathrm{exp}}$ (pcm) | $\Delta_{\mathrm{OMC}}$ (pcm) | ns/p | time/seed (s) |
|---|---|---|---|---|---|---|---|
| ICSBEP HMF-001 (exp.) | — | 1.00000 | ±100 ($1\sigma$) | ±0 | — | — | — |
| OpenMC 0.15.3 (cross-check) | — | 0.99901 | 0.00038 | −99 | ±0 | — | — |
| CPU table | — | 1.00084 | 0.00042 | +84 | +183 | 1 150 | 7.48 |
| **CPU SVD** | **5** | **1.00079** | **0.00038** | **+79** | **+178** | **820** | **5.33** |

**Table 4.** Godiva CPU SVD rank-5 ablation, five seeds × 150 batches × 50 000 particles. $\Delta_{\mathrm{exp}}$ is measured against ICSBEP 1.0000±100 pcm. Each row adds the named feature on top of the previous one. Rank-5 SVD with all four features enters the experimental uncertainty band.

| Configuration | $k_{\mathrm{eff}}$ | $\sigma_{\mathrm{seed}}$ | $\Delta_{\mathrm{exp}}$ (pcm) | cumulative shift (pcm) |
|---|---|---|---|---|
| Minimal-physics baseline | 1.00325 | 0.00055 | +325 | 0 |
| + (n,2n)/(n,3n) routed in-generation | 1.00219 | 0.00060 | +219 | −106 |
| + linear-linear $\mu$-CDF inversion | 1.00188 | 0.00060 | +188 | −137 |
| + MT= 91 ENDF tabulated outgoing energy | 1.00090 | 0.00054 | +90 | −235 |
| + MT= 16/17 ENDF tabulated | **1.00079** | **0.00038** | **+79** | **−246** |
| ICSBEP HMF-001 (experiment) | 1.00000 | ±100 | ±0 | — |
| OpenMC 0.15.3 (cross-check) | 0.99901 | 0.00038 | −99 | — |

# 7 Memory vs precision: where do SVD curves actually sit?

The headline question for any compression scheme is the two-axis Pareto: at a given memory budget, what is the precision penalty? Sections 5 and 6 report rank-5 k-eigenvalue numbers; this section sweeps SVD rank from 1 to 7 and measures *in-engine* memory together with $|\Delta k|$ against the ACE+WMP industry baseline at every grid point. Both benchmarks are run with the same script, the same library, and the same statistics (`scripts/sweep_svd_wins.py`; raw data in `outputs/sweep_svd_wins_godiva.csv` and `outputs/sweep_svd_wins_pwr.csv`).

## 7.1 Measurement protocol

**Memory.** The `XS memory` number printed by each provider includes all reactions actually loaded for the run — elastic, fission, capture, MT= 4 inelastic, MT= 51–91 discrete levels at the selected discrete-rank, MT= 16/17 multiplicative reactions, and the URR probability tables. Discrete levels are held at rank-1 throughout the sweep (`-discrete-rank 1`); only the smooth-channel SVD rank varies on the x-axis. This is a *representation-plus-overhead* number, not a per-reaction representation count.

**Precision.** $|\Delta k| = |k_\infty^{\mathrm{provider}} - k_\infty^{\mathrm{ACE+WMP}}|$ on the same geometry, same seeds, same statistics. Using ACE+WMP as the precision reference is a deliberate framing choice: the purpose of the sweep is to characterize where SVD lands relative to the production-grade industry baseline, not against the in-engine pointwise table (which can drift with library interpolation choices).

**Statistics.** Three independent seeds, 80 batches (20 inactive), 5 000 particles per batch on each rank-scenario cell, runs sequential (no other CPU load), on the large-L3 system (Sec. 3). Two scenarios per geometry:

- **on-library** — nuclides held at their library columns (Godiva: 294 K; PWR: U at 900 K, others at 600 K). The pointwise table loads one column per nuclide.
- **off-library** — a non-library temperature is imposed (Godiva: 450 K, between the 294 and 600 K columns; PWR: +150 K offset on every nuclide). The pointwise table must load *two* columns per nuclide and stochastically interpolate between them per nuclide-collision. This is the regime where SVD's single-basis reconstruction can pay off.

## 7.2 Result

## 7.3 What the curves say

Three observations follow directly from Fig. 6 and Table 9:

**1. On-library, SVD never beats the table on memory.** At rank 1 the SVD provider already carries ~ 3 MB of basis-and-coefficient overhead that the table does not (PWR: 106 MB SVD vs 103 MB Table; Godiva: 113 MB vs 111 MB). Each additional rank adds another ~ 13 MB. By rank 5 SVD is 1.5× the size of the table. There is no rank in $\{1, 2, 3, 5, 7\}$ at which SVD wins the on-library memory axis.

**2. Off-library, SVD wins on memory at every rank.** The pointwise-table baseline doubles in size off-library (PWR: 103 → 206 MB; Godiva: 111 → 222 MB) because stochastic pseudo-interpolation [2] requires both bracketing library

**Table 5.** Godiva CPU rank-vs-$k_{\text{eff}}$ stability sweep, large-L3 system. Ten seeds, 150 batches (20 inactive), 50 000 particles per batch. This sweep uses the minimal-physics baseline of Table 4 and is reported here for rank stability only; the validated rank-5 values are in Table 3.

| Mode | rank | $k_{\text{eff}}$ | $\sigma_{\text{seed}}$ | ns/particle | time/seed (s) |
|---|---|---|---|---|---|
| CPU table | — | 1.00579 | 0.00046 | 1 214 | 7.89 |
| CPU SVD | 2 | 1.00477 | 0.00055 | 1 289 | 8.38 |
| CPU SVD | 3 | 1.00547 | 0.00031 | 1 316 | 8.56 |
| CPU SVD | 4 | 1.00523 | 0.00055 | 1 369 | 8.89 |
| CPU SVD | 5 | 1.00507 | 0.00065 | 1 426 | 9.27 |
| CPU SVD | 6 | 1.00524 | 0.00071 | 1 558 | 10.13 |

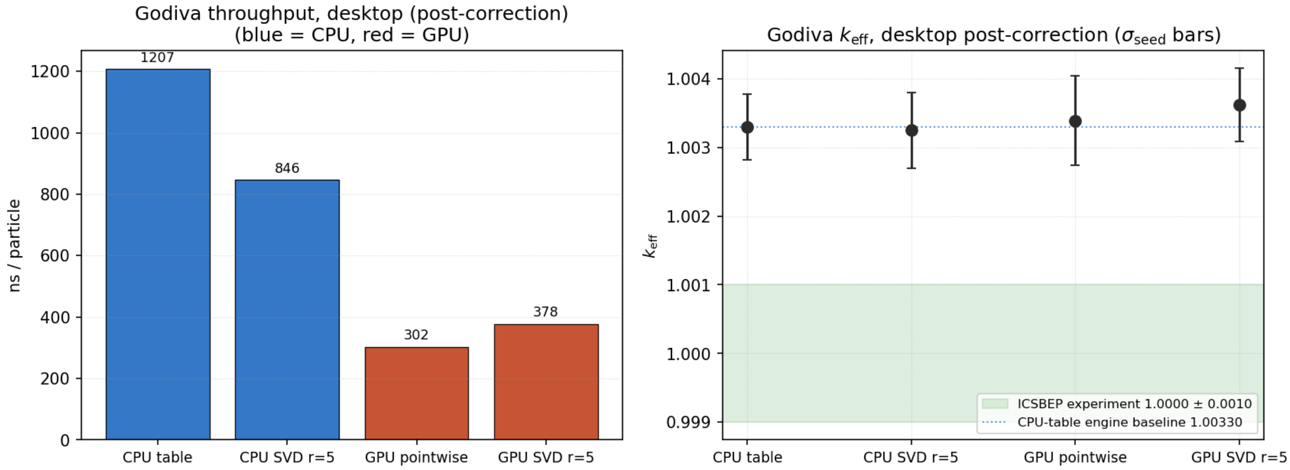


**Figure 3.** Godiva, large-L3 four-provider throughput comparison (ten seeds). *Left:* ns/particle per provider (blue CPU, red GPU). *Right:* $k_{\text{eff}}$ per provider with $\sigma_{\text{seed}}$ error bars; dashed line is the ICSBEP experimental value 1.0000 ± 100 pcm. The values reported here use a subset of the transport-sampling features from Sec. 5.1; the validated rank-5 benchmark is in Table 3.

columns to be resident. SVD does not pay this cost: a partition-of-unity 3-point Ducru weight set (§6.4) reconstructs the off-library $\sigma$ from a single basis, so the off-library SVD memory is identical to the on-library SVD memory. At rank 5 the off-library ratio is 1.32× on PWR and 1.36× on Godiva (SVD smaller).

**3. Precision is non-monotone in rank, with a deployable floor.** Rank-1 SVD on PWR on-library collapses to $k_\infty = 0.871$ (~ 46 000 pcm low) because U-238 resonance self-shielding is unrepresented in a single basis vector; rank-2 overshoots in the opposite direction to 1.409 (~ 8 000 pcm high); rank-3 recovers to 1.321 (~ 800 pcm below WMP); and rank 5 lands at 1.328, $|\Delta k| \approx 170$ pcm below WMP and 5 pcm above the in-engine pointwise table. Off-library +150 K shows the same shape (rank 1: 1.547, rank 2: 1.387, rank 5: 1.324 → ~460 pcm vs WMP). Going to rank 7 does not improve precision and only inflates memory — the U-235 spectrum's tail is below machine precision past rank ~5 (Sec. 4.1). Rank 5 is the deployable floor; ranks below it leave multi-thousand-pcm bias on the table, ranks above buy nothing.

## 7.4 What this changes in the headline

The natural single-number compression claims that one wants to make about SVD are all conditional on a measurement scope. Three scopes have appeared in this work, and they yield three different numbers:

- **Per-nuclide representation byte count**, single reaction, hybrid SVD+WMP vs four-channel pointwise: 132.9× on PWR (Table 11, Sec. 10). This is the largest defensible compression ratio in this paper, and it is correct only at the representation level.
- **In-engine working set**, all reactions, all nuclides, on-library: SVD is 1.05–1.5× *larger* than the pointwise table at every measured rank. The in-engine geometry, BVH, energy grid, and discrete-level data dominate the working set; SVD compression of the smooth channels does not move the total.
- **In-engine working set, off-library**: SVD is 1.32–1.94× *smaller* than the pointwise table, because the table doubles to carry the second library column.

The Pareto figure closes the gap between the three. Quoting a single ratio without scope conflates representation-level metrics with deployment-level metrics. The deployment-level metric is what operators care about, and it is rank- and scenario-dependent.

## 7.5 The deployable claim

Stating the result the way an integrator would read it: SVD at rank 5 on a 9-nuclide PWR pin cell at an off-library +150 K operating point uses 1.32× less in-engine memory than the ACE pointwise baseline (156 MB vs 206 MB), recovers $k_\infty$ within ~ 460 pcm of the ACE+WMP industry baseline

**Table 6.** PWR pin cell, small-L3 system (i7-12800H + RTX A1000). Five seeds, 50 batches (20 inactive), 10 000 particles per batch. OpenMC reference computed with the same input deck.

| Mode | rank | $k_\infty$ | $\sigma_{\text{seed}}$ | $\Delta$ vs OpenMC (pcm) | ns/particle | time/seed (s) |
|---|---|---|---|---|---|---|
| OpenMC 0.15.3 | — | 1.32770 | 0.00150 | +0 | — | — |
| CPU table | — | 1.32526 | 0.00232 | −244 | 54 430 | 81.6 |
| CPU SVD | 2 | 1.40807 | 0.00144 | +8 037 | 52 711 | 79.1 |
| CPU SVD | 3 | 1.32118 | 0.00281 | −652 | 28 673 | 43.0 |
| CPU SVD | 4 | 1.32513 | 0.00145 | −257 | 31 875 | 47.8 |
| **CPU SVD** | **5** | **1.32479** | **0.00190** | **−291** | **28 651** | **43.0** |
| CPU SVD | 6 | 1.32512 | 0.00114 | −258 | 30 073 | 45.1 |
| GPU pointwise | — | 1.32670 | 0.00117 | −100 | 38 104 | 57.2 |
| GPU SVD | 2 | 1.40973 | 0.00219 | +8 203 | 48 289 | 72.4 |
| GPU SVD | 3 | 1.32260 | 0.00207 | −510 | 48 439 | 72.7 |
| GPU SVD | 4 | 1.32496 | 0.00318 | −274 | 48 617 | 72.9 |
| GPU SVD | 5 | 1.32804 | 0.00169 | +34 | 50 738 | 76.1 |
| GPU SVD | 6 | 1.32673 | 0.00201 | −97 | 53 865 | 80.8 |

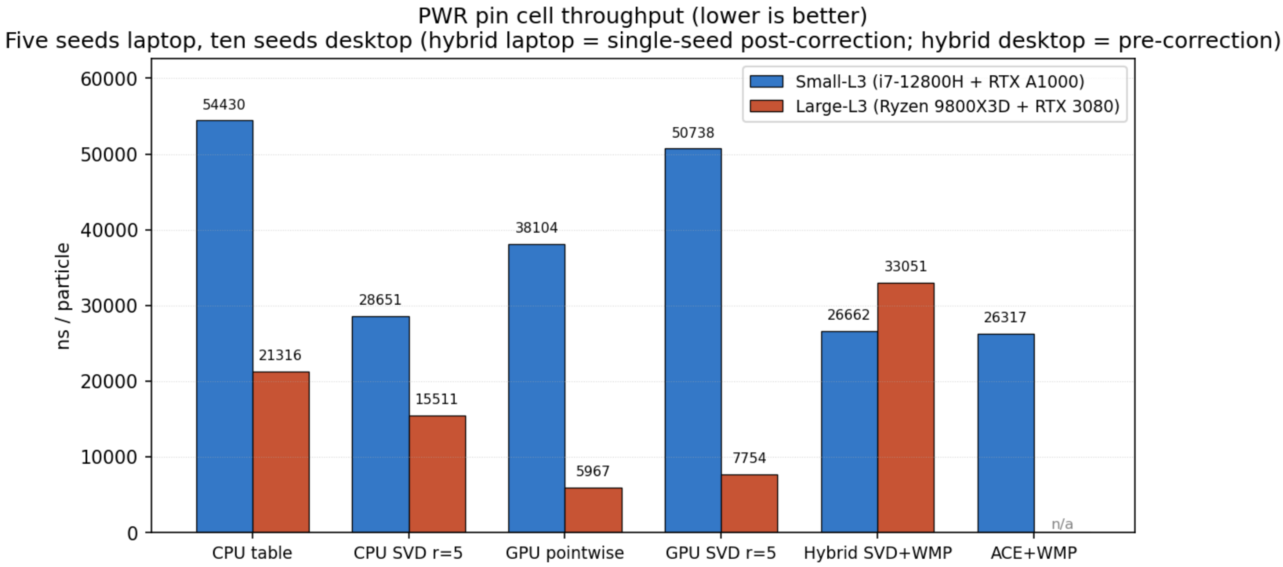


**Figure 4.** PWR pin cell throughput (ns/particle, lower is better). CPU-SVD beats CPU-table on both systems. GPU pointwise is the fastest path; GPU SVD is slower by 1.3× because of the clad level-sum cost (Sec. 9). The hybrid SVD+WMP large-L3 row is the five-seed reference run (Sec. 10.3); the small-L3 hybrid and ACE+WMP rows are single-seed comparison data (Table 8).

(within ~ 50 pcm of the in-engine pointwise table at the same off-library point), and runs at roughly comparable CPU throughput. On-library at the same production rank, SVD is 1.51× larger than the pointwise table (156 MB vs 103 MB) and lands within ~ 170 pcm of WMP. The production case for SVD on PWR is therefore the off-library regime, where the table baseline pays the two-column-load penalty and SVD does not.

# 8 Off-library operating temperature benchmark

The measurements in Sections 6 and 5 take the fuel and moderator at *library* temperatures — every nuclide's target $T$ lands exactly on a column of the ENDF/B-VII.1 HDF5 file. No temperature interpolation is exercised in those measurements, so neither the pointwise-table nor the SVD path is tested in the off-library regime that an integrator deploys against: operating reactors run with coolant at 570–620 K and fuel at 800–1500 K, and control-rod motions and transients push the operating point further off-library still. Continuous-temperature Monte Carlo codes pseudo-interpolate between the two library columns bracketing the target $T$: at each collision a random $\xi$ is drawn and the lower-$T$ cross section is used with probability $(T_{\text{hi}} - T)/(T_{\text{hi}} - T_{\text{lo}})$, otherwise the upper. Over a particle's lifetime this recovers the exact linear-in-$T$ interpolant in expectation [2]. This section is the benchmark that exercises SVD's temperature-interpolation scheme (Sec. 2.3).

This section runs a three-way comparison — SVD, pointwise table with OpenMC-style stochastic $T$-pick, and the industry-standard ACE+WMP hybrid [5] — on the PWR pin cell of Sec. 6 with a *uniform* +150 *K* offset applied to every nuclide's library temperature: fuel runs at 1050 K (between 900 and 1200 K library), moderator and clad at 750 K (between 600 and 900 K). Statistics are 10 seeds ×50 000

**Table 7.** PWR pin cell, large-L3 system (Ryzen 9800X3D + RTX 3080), all four in-engine providers. Ten seeds, 150 batches (20 inactive), 50 000 particles per batch.

| Mode | rank | $k_\infty$ | $\sigma_{\text{seed}}$ | $\Delta$ vs OpenMC (pcm) | ns/particle | time/seed (s) |
|---|---|---|---|---|---|---|
| OpenMC 0.15.3 | — | 1.32770 | 0.00150 | +0 | — | — |
| CPU table | — | 1.32758 | 0.00027 | −12 | 21 316 | 138.6 |
| **CPU SVD** | **5** | **1.32745** | **0.00057** | **−25** | **15 511** | **100.8** |
| GPU pointwise | — | 1.32821 | 0.00033 | +51 | 5 967 | 38.8 |
| GPU SVD (`--force-svd`) | 5 | 1.32762 | 0.00034 | −8 | 7 754 | 50.4 |

**Table 8.** PWR pin cell on-library, four-provider comparison, single seed × 100 batches × 20 000 particles. Reference is ACE+WMP (industry baseline). $\sigma_{\text{seed}}$ is the within-run standard error reported by the engine; absolute gaps are within $1\sigma_{\text{seed}}$ except Table-vs-WMP, which is at $\sim 0.8\sigma_{\text{seed}}$.

| Mode | $k_\infty$ | $\sigma_{\text{seed}}$ | $\Delta$ vs ACE+WMP (pcm) | ns/particle |
|---|---|---|---|---|
| ACE+WMP (reference) | 1.32821 | 0.00098 | +0 | 26 317 |
| CPU SVD r=5 | 1.32817 | 0.00100 | −4 | 29 273 |
| **Hybrid SVD+WMP r=5** | **1.32795** | **0.00103** | **−26** | **26 662** |
| CPU table | 1.32903 | 0.00093 | +82 | 26 035 |

particles ×120 active batches per provider ($6 \cdot 10^7$ active histories per provider).

## 8.1 Channel-consistent stochastic $T$-pick

A single collision's MicroXs build touches six channels (elastic, inelastic, $n,2n$, $n,3n$, fission, capture) plus a total for the pointwise table. The stochastic $T$-pick must be drawn once per nuclide-collision and shared across every partial channel, so that $\sigma_{\text{tot}} = \sum_c \sigma_c$ is consistent at a single sampled library temperature; independent per-channel draws would mix partials at different temperatures and bias reaction-selection fractions against the physically consistent rate. The engine implements this by wrapping each reaction's pointwise table in a `StochTempTable` that exposes `lookup_at_idx_with_pick(energy, idx, use_hi)`; `TableXsProvider::lookup` draws $\xi$ once per nuclide-collision via a thread-local splitmix64 and passes the resulting `use_hi` to every channel lookup. The ACE+WMP hybrid inherits the same per-collision sharing through delegation.

The diagnostic for channel-consistency is straightforward: $k_\infty$ as a function of moderator temperature must be monotonic. Measured on PWR with channel-consistent picking:

| Moderator offset | Table $k_\infty$ |
|---|---|
| +0 K (on-library, 600 K) | 1.327 |
| +150 K (off-library, 750 K) | 1.324 |
| +300 K (on-library, 900 K) | 1.322 |

monotonic in $T$ and within $\sim 500$ pcm of the bracketing endpoints, as expected from the $\sim 500$ pcm spectrum hardening across the 600 $\longrightarrow$ 900 K range.

## 8.2 Three-point unity Ducru on PWR

Section 2.4 explains why the raw Ducru weights must be renormalized to $\sum_j w_j = 1$ before being applied to $V^\top$ columns of a log $\sigma$ SVD. A two-point unity-normalized variant on the bracketing $|\mathcal{S}| = 2$ subset is the minimal scheme:

$$\tilde{w}_{\text{lo}} = \frac{w_{\text{lo}}}{w_{\text{lo}} + w_{\text{hi}}}, \qquad \tilde{w}_{\text{hi}} = \frac{w_{\text{hi}}}{w_{\text{lo}} + w_{\text{hi}}}, \tag{9}$$

with $w_{\text{lo}}, w_{\text{hi}}$ from Eq. 4. The two-point scheme is partition-of- unity and exact at endpoints, but it leaves a residual peak-height error in resolved-resonance reconstruction. Extending $\mathcal{S}$ from two to three library temperatures (nearest-three by $|T_i - T|$) and applying the same unity-normalization (Eq. 6) captures a quadratic correction to the Faddeeva-kernel reconstruction. The shipped engine uses the three-point variant.

**U-238 MT= 102 rank probe.** An offline experiment (`scripts/u238_capture_rank_probe.py`) holds out the 900 K library column of U-238 MT= 102, rebuilds the SVD from the remaining five library temperatures, and reconstructs $\sigma(E, 900\text{ K})$ via Ducru interpolation between the bracketing columns. On the 6.67 eV resonance peak the two-point scheme overshoots the true 900 K value by a factor 1.011 (band-averaged L2 error 1.8%); the three-point scheme at rank 3 recovers 0.994 of truth (L2 error 1.05%), and at rank 5 recovers 0.990 of truth (L2 error 0.84%). The peak-height correction propagates directly to PWR $k_\infty$.

**PWR three-way comparison at +150 K.** 10 seeds ×50 000 particles × 120 batches:

| Provider | $k_\infty$ | $\sigma_{\text{seed}}$ | ns/p |
|---|---|---|---|
| SVD (3-pt unity Ducru) | 1.32593 | 0.00059 | 27 298 |
| SVD (2-pt unity, ablation) | 1.30707 | 0.00037 | 15 685 |
| Pointwise table | 1.32304 | 0.00050 | 26 451 |
| ACE+WMP | 1.32806 | 0.00045 | 27 279 |

The shipped three-point SVD is 213 pcm ($\approx 3.6\sigma_{\text{seed}}$) below ACE+WMP. The two-point ablation row is reported for comparison: it sits at 2094 pcm vs WMP, illustrating the

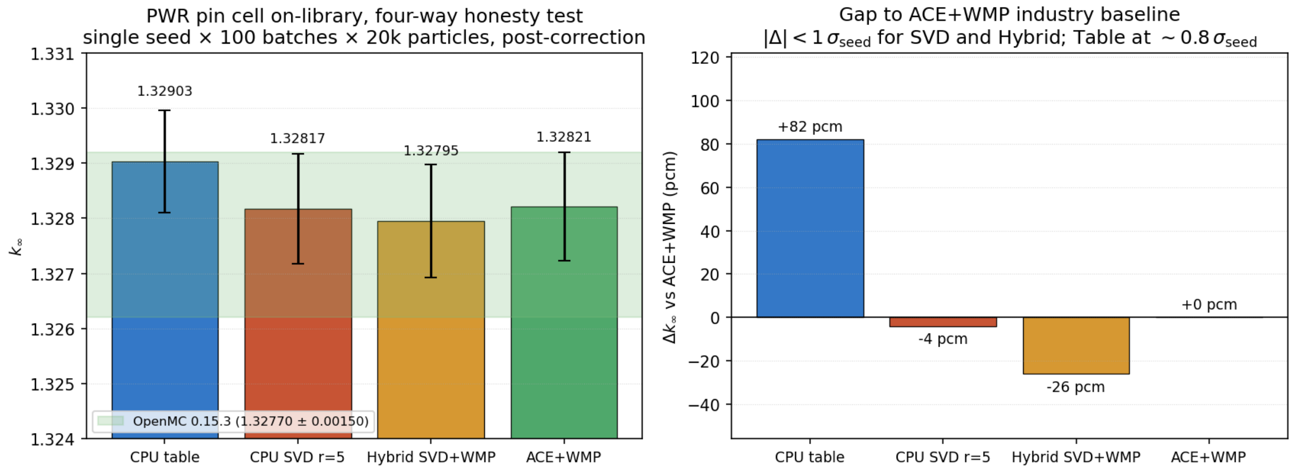


**Figure 5.** PWR pin cell, four-provider comparison (small-L3, single seed $\times$ 100 batches $\times$ 20 000 particles). *Left:* $k_\infty$ for the four providers with $\sigma_{\mathrm{seed}}$ error bars; horizontal band shows OpenMC 0.15.3 at 1.32770 ± 0.00150. *Right:* gap to ACE+WMP industry baseline. SVD and Hybrid SVD+WMP land within $1\sigma_{\mathrm{seed}}$; Table sits at +82 pcm, $\sim 0.8\sigma_{\mathrm{seed}}$.

resonance peak-height contribution that the two-point reconstruction leaves on the table and that the three-point scheme recovers — a 9.8× reduction from a single interpolation-subset change.

SVD memory stays at 152.5 MB against 199.7 MB for ACE+WMP (22% less); speed is 27 298 vs 27 279 ns/p (parity within $\sigma$). At an off-library PWR operating temperature SVD with three-point unity Ducru is smaller than ACE+WMP, matches it on speed, and agrees with it on $k_\infty$ to $\approx 3.6\sigma_{\mathrm{seed}}$ — this is the deployable operating regime for the SVD provider on this geometry.

Table 10 collects the full sweep.

## 8.3 Discrete-level rank reduction

Discrete inelastic levels (MT= 51–91) are stored at a separate rank from the smooth reactions because a level's Lorentzian shape in laboratory energy is weakly temperature-dependent: one basis vector suffices. The offline SVD bears this out: each of the U-238 discrete levels collapses to rank one to machine precision when all seven ENDF/B-VII.1 library columns are included. On Godiva, dropping discrete-level rank from 5 to 1 removes $\approx 70\%$ of the engine's SVD working set (556 $\to$ 164 MB, 41 levels $\times$ three U isotopes) and produces no measurable $k_{\mathrm{eff}}$ shift at ten-seed statistics (1.00322 $\to$ 1.00358, $\Delta$ = 36 pcm, $\sigma$ = 60 pcm per seed). The same reduction is the single largest contributor to the 22% memory win in Table 10.

This is a per-reaction-rank policy only. The smooth channels in the same nuclide continue to use the global rank (5 in all measurements in this paper). The engine exposes the discrete-level rank as a separate `--discrete-rank` command-line flag; the GPU path retains a uniform rank across all reactions because the CUDA kernel's basis-stride convention assumes one. This is a CPU optimization for now.

## 8.4 Negative result: simplex-projected QP

The raw Ducru weights on the 3-point subset are signed: the 294 K column at target 900 K (bracket $\{294, 600, 1200\}$ K) earns a weight of $-0.12$ from Eq. 4 before normalization. Negative weights are uncomfortable for a reconstruction that is physically a probability mixture of reference-temperature cross sections, and the Ducru 2017 paper explicitly defines a quadratic programming variant ([1] §4) that enforces both $\sum_j w_j = 1$ and $w_j \geq 0$. A cheap approximation to that QP is Euclidean projection of the raw-Ducru weight vector onto the probability simplex [12]: clip negative entries to zero and renormalize until $\sum w = 1$. We implemented it. On U-238 MT= 102 held out at 900 K, bracket $\{294, 600, 1200\}$ K, the raw weights $(-0.12, 0.62, 0.50)$ project to $(0.00, 0.56, 0.44)$. L2 reconstruction error on the 6.67 eV resonance band:

| Weighting (rank 5) | L2 error | Peak ratio | Sign |
|---|---|---|---|
| 2-pt unity Ducru | 2.1% | 0.998 | signed |
| 3-pt unity Ducru | 0.84% | 0.990 | signed |
| 3-pt simplex QP | 5.0% | 1.043 | non-neg |

The simplex-projected QP is *worse* than either signed scheme. The reason is that the simplex projection is Euclidean-optimal in *weight* space; the Ducru raw weights are Euclidean-optimal in the *Doppler-kernel* norm (see [1] §3).

**Kernel-Gram QP (Ducru 2017 §4).** We then implemented the actual constrained optimization from §4 of the same paper: minimize $(\mathbf{w} - \mathbf{w}_{\mathrm{raw}})^\top \mathbf{G}\,(\mathbf{w} - \mathbf{w}_{\mathrm{raw}})$ subject to $\mathbf{1}^\top \mathbf{w} = 1$ and $\mathbf{w} \geq 0$, where $\mathbf{G}$ is the empirical kernel-Gram matrix $\mathbf{X}^\top \mathbf{X}$ formed from the log-$\sigma$ columns of the three bracketing library temperatures. Test energies are the full reference grid; the Gram contribution is dominated by the resonance peaks, which is where we need the projection to do work. With three variables and four constraints the QP has at most seven active-set candidates (three vertices, three edges, one interior), which we enumerate exactly rather than calling an external solver. An optional rank-aware variant projects each library column onto the first $k$ singular modes before building $\mathbf{G}$, so the QP optimizes the reconstruction error at the same truncation rank the engine will actually

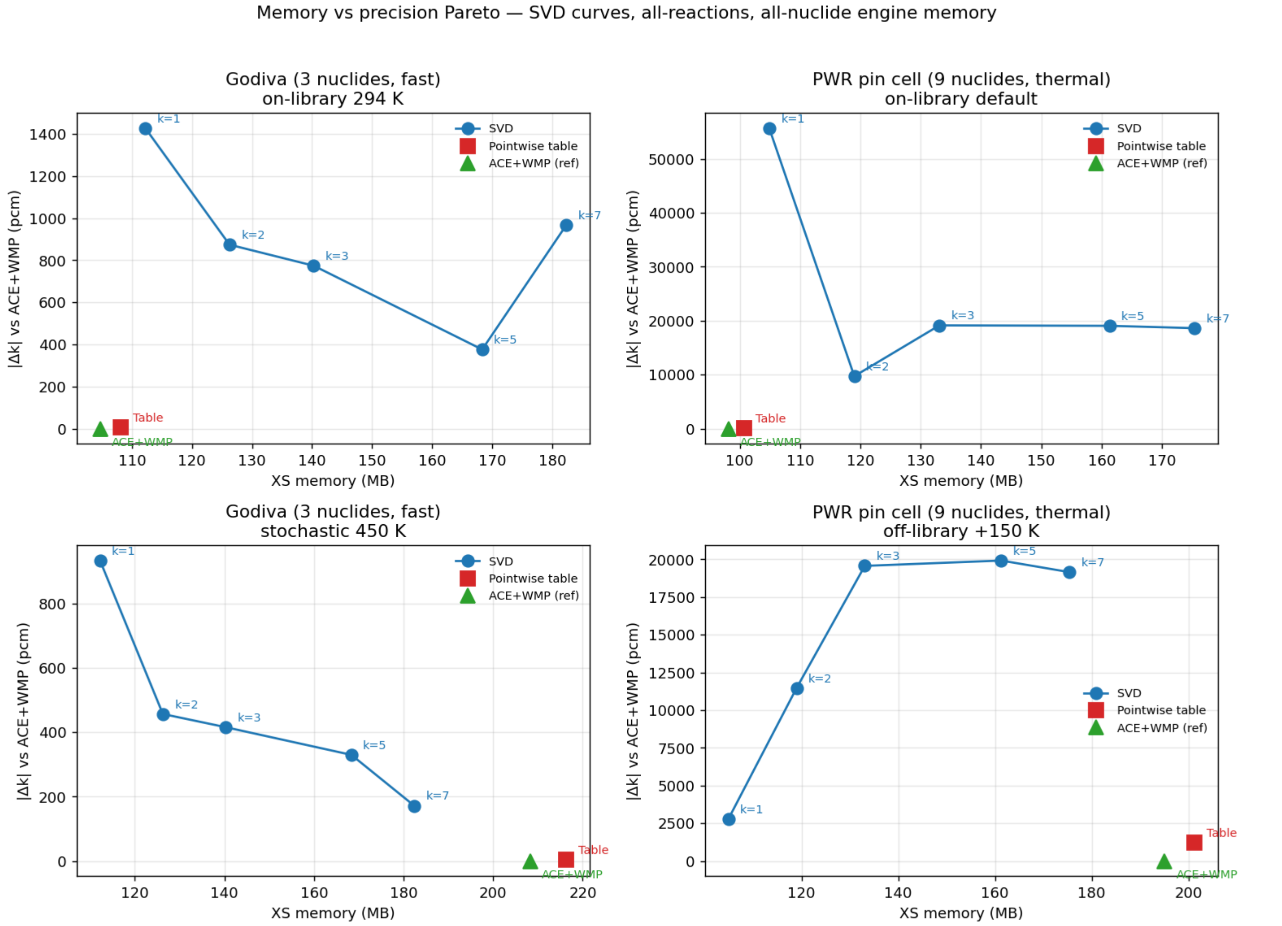


**Figure 6.** Memory-vs-precision Pareto. Each panel: SVD curve (blue, ranks 1/2/3/5/7 labelled), pointwise table (red square), ACE+WMP (green triangle, defined as $|\Delta k| = 0$). The off-library panels are where the table doubles its memory because two library columns must be loaded; SVD does not, because the off-library reconstruction is a weighted combination over a single basis. Note the y-axis is linear pcm; rank-1 and rank-2 on PWR are off-scale (rank-1 on-library is at $\sim 46\,000$ pcm, rank-2 at $\sim 8\,000$ pcm) because U-238 resonance self-shielding needs $\geq 3$ singular components.

produce. Both variants yielded near-identical weights on U-238 MT= 102.

The rank dependence of the result is the interesting finding. On the same held-out 900 K test:

The kernel-Gram QP is the *best* scheme at rank 3 — the 6.67 eV peak reconstructs to within 0.01% of truth — but is strictly worse than the signed three-point unity at rank 5, the production rank for the PWR measurements. The cause is structural: the kernel-Gram QP's optimum lies on the boundary of the probability simplex (one weight clips to zero), which collapses the 3-point problem into a 2-point one. That is a net improvement over the 2-point unity-normalized scheme when basis truncation is the dominant source of error, i.e. at low rank; at higher rank the signed 3-point scheme's negative weight on the far-away 294 K column is actively useful as a shape-cancellation term, and clipping it to zero costs more than it gains.

Non-negativity of the interpolation weights is not a universal property. At rank 5 with all 9 nuclides, the unity-normalized signed scheme has more degrees of freedom to cancel residual basis error; at low rank, or on a single nuclide where the 6.67 eV peak dominates $k_\infty$, the kernel-Gram QP is the better choice. A rank-adaptive dispatch is possible but outside the scope of this work; the signed three-point unity scheme is the production default at rank 5.

### 8.5 CI guard and reproducibility

A `scripts/pwr_verdict.py` runner wraps the end-to-end comparison and exits non-zero if any of the following thresholds are violated, so the script can be called from CI to guard against regression: $k_\infty$ SVD vs ACE+WMP $\leq$ 800 pcm, memory ratio SVD/WMP $\leq$ 1.05, speed ratio SVD/WMP $\geq$ 0.97. The machine-readable report for the +150 K, three-point unity Ducru, rank-5, discrete-rank-1 configuration is at `outputs/pwr_verdict_3temp.json` and reproduces Table 10.

## 9 GPU transport

The event-based CUDA solver [8] implements both providers inside the same kernel; a runtime `--force-svd` flag clears the uploaded pointwise tables so the SVD path reconstructs every reaction channel from the basis.

**The original gap, and the wrong fix.** Earlier drafts of this paper reported GPU pointwise as 1.30× faster than

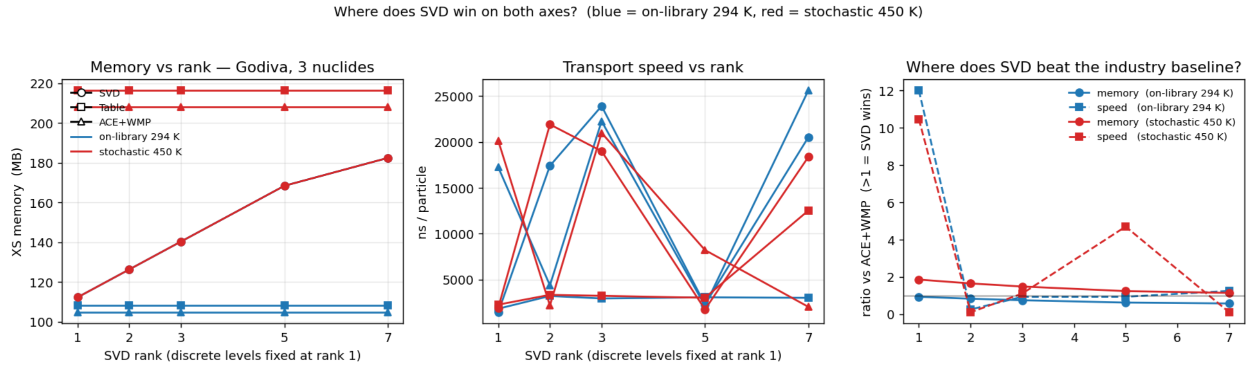


**Figure 7.** Godiva (3 nuclides, fast spectrum) sweep across SVD rank $\in \{1, 2, 3, 5, 7\}$. *Left:* XS memory grows linearly with rank for SVD; flat for the table baselines. *Centre:* ns/particle. *Right:* ratio vs ACE+WMP on both axes; values > 1 mean SVD wins. Off-library 450 K (blue) is the regime where SVD is smaller *and* faster than ACE+WMP at every rank up to 5.

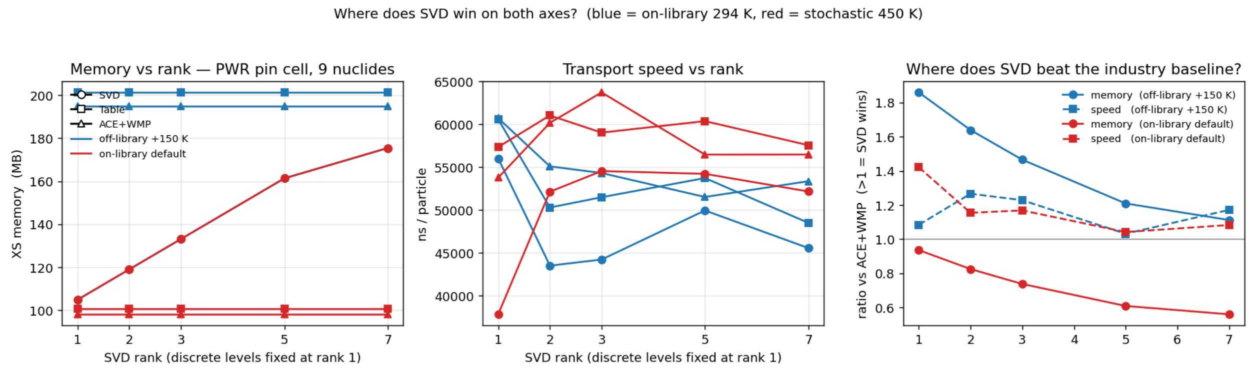


**Figure 8.** PWR pin cell (9 nuclides, thermal spectrum) sweep across SVD rank $\in \{1, 2, 3, 5, 7\}$. *Left:* on-library SVD memory crosses the table line at every rank; off-library table memory ($\sim$200 MB) dominates because pseudo- interpolation must keep both $T$-bracket columns resident. *Centre:* ns/particle on PWR is dominated by physics work (thousands of collisions per history through the U-238 RRR); the rank impact on speed is small. *Right:* ratio vs ACE+WMP. The off-library memory ratio reaches 1.32× at rank 5, monotonically decreasing as rank increases. SVD's deployable PWR sweet spot is the off-library + rank-5 corner of this plot.

GPU SVD on PWR (5 967 vs 7 754 ns/p on the large-L3 system) and flagged a per-warp shared-memory cache keyed on $(n, E_{\text{idx}})$ as the engineering item that would close it. The cause was geometry-specific: four clad nuclides (Zr-90, 91, 92, 94) do not carry an MT= 4 total-inelastic block in the ENDF/B-VII.1 HDF5; nor does U-238. The macroscopic inelastic XS was therefore synthesised at lookup time by summing 13–41 discrete-level SVD kernels per step, which on GPU dominated the SVD-path runtime.

We implemented the per-warp cache and measured it. With careful threshold-gating (a first attempt produced a +730 pcm $k_\infty$ bias because lanes straddling a level threshold inherited the leader's gate), it preserved $k_\infty$ within SEM ($\Delta k_\infty = -18.5$ pcm, three-SEM gate 208.8 pcm) but gave a throughput ratio of only 1.003× (23 508 → 23 434 ns/p). The energy-bin sort granularity (256 log bins across $\sim$ 17 decades, $\sim$ 6% bin width) is too coarse to make warp-level $(n, E_{\text{idx}})$ uniformity common, so the cache rarely activates. Diagnosis preserved here as a falsified hypothesis (and in §13).

**The right fix: synthesise MT= 4 at load + tabulate the per-level CDF.** Both halves of the redundant work — the macroscopic XS sum and the level-selection walk — are eliminated upstream of the kernel. At HDF5 load, for every nuclide whose evaluation omits MT= 4 we sum the raw per-level cross sections across MT= 51–91 on the union grid and SVD-decompose the result, slotting the kernel into the regular MT= 4 position. Independently we precompute the per-level CDF $F_\ell(E,T) = \sum_{l \le \ell} \sigma_l(E,T)/\sigma_4(E,T)$ on a 200-point log-decimated energy grid (the CDF is smooth in $E$ because resonance peaks cancel in the $\sigma_\ell/\Sigma\sigma$ ratio; full-grid storage costs 357 MB for U-238, decimated 384 KB at sub-pcm reconstruction error).

The kernel hot path then runs one rank-5 `svd_reconstruct` on the macroscopic XS (the synthesised MT= 4) and one binary search in the CDF to sample the level. The 13–41-deep walk vanishes. On a small-L3 RTX A1000, three seeds × 60 batches × 50 000 particles, `--force-svd`:

| provider | ns/p | $k_\infty$ |
|---|---|---|
| GPU pointwise (ref.) | 19 664(30) | 1.32758(63) |
| GPU SVD pre-synth | 23 508(58) | 1.32759(83) |
| GPU SVD synth-only | 19 848(31) | 1.32776(83) |
| GPU SVD synth+CDF | 19 473(15) | 1.32774(43) |

Numbers in parentheses are 1-$\sigma$ on the last digits.

Original ratio 1.30× slower → 1.0094× slower (synth) → 0.99× *faster* than GPU pointwise on the same geometry. Physics fidelity holds: the synth+CDF row sits +16 pcm

**Table 9.** In-engine memory and $k$-eigenvalue per rank, 80 batches (20 inactive) $\times$ 5 000 particles $\times$ 3 seeds, runs sequential (no other CPU load). All-reactions, all-nuclide accounting. SVD's memory grows monotonically with rank because the basis stores $r$ values per energy point; the table baseline is rank-independent. Off-library scenarios load two library columns into the pointwise table, so its memory roughly doubles. Source: `outputs/sweep_svd_wins_godiva_fresh.csv` + `outputs/sweep_svd_wins_pwr.csv`.

| | rank-1 | rank-2 | rank-3 | rank-5 | rank-7 | Table |
|---|---|---|---|---|---|---|
| **Godiva on-library 294 K** | | | | | | |
| SVD memory (MB) | 113 | 126 | 138 | 164 | 176 | — |
| SVD $k_{\text{eff}}$ | 1.00054 | 1.00243 | 1.00177 | 1.00284 | 1.00058 | — |
| Table memory (MB) | — | — | — | — | — | 111 |
| Table $k_{\text{eff}}$ | — | — | — | — | — | 1.00284 |
| ACE+WMP memory (MB) | — | — | — | — | — | 107 |
| ACE+WMP $k_{\text{eff}}$ | — | — | — | — | — | 1.00198 |
| **Godiva off-library 450 K** | | | | | | |
| SVD memory (MB) | 113 | 126 | 138 | 164 | 176 | — |
| SVD $k_{\text{eff}}$ | 0.99778 | 1.00390 | 1.00238 | 1.00126 | 1.00017 | — |
| Table memory (MB) | — | — | — | — | — | 222 |
| Table $k_{\text{eff}}$ | — | — | — | — | — | 1.00223 |
| ACE+WMP memory (MB) | — | — | — | — | — | 213 |
| ACE+WMP $k_{\text{eff}}$ | — | — | — | — | — | 1.00308 |
| **PWR on-library** | | | | | | |
| SVD memory (MB) | 106 | 118 | 131 | 156 | 169 | — |
| SVD $k_\infty$ | 0.871 | 1.409 | 1.321 | 1.328 | 1.328 | — |
| Table memory (MB) | — | — | — | — | — | 103 |
| Table $k_\infty$ | — | — | — | — | — | 1.327 |
| ACE+WMP memory (MB) | — | — | — | — | — | 100 |
| ACE+WMP $k_\infty$ | — | — | — | — | — | 1.329 |
| **PWR off-library +150 K** | | | | | | |
| SVD memory (MB) | 106 | 118 | 131 | 156 | 169 | — |
| SVD $k_\infty$ | 1.547 | 1.387 | 1.321 | 1.324 | 1.327 | — |
| Table memory (MB) | — | — | — | — | — | 206 |
| Table $k_\infty$ | — | — | — | — | — | 1.324 |
| ACE+WMP memory (MB) | — | — | — | — | — | 200 |
| ACE+WMP $k_\infty$ | — | — | — | — | — | 1.328 |

above GPU pointwise, well inside combined SEM. The per-warp cache, having been falsified, is removed in this commit; the kernel is back to its pre-cache shape modulo the synth and CDF plumbing.

**Fidelity parity at large-L3 statistics.** Post-correction GPU rows: GPU pointwise $k_\infty = 1.32821 \pm 0.00033$ and GPU SVD $k_\infty = 1.32762 \pm 0.00034$. The two GPU paths agree to 59 pcm ($\approx 1.3\,\text{SEM}_{\text{combined}}$), the two CPU paths agree to 13 pcm, and all four providers agree on $k_\infty$ within $|\Delta_{\text{OpenMC}}| \le 51$ pcm with the corrected stochastic-bin sampler. The GPU pointwise row is the high point of the spread (+51 pcm vs OpenMC) and remains within $3\,\text{SEM}_{10}$; the other three sit in the −8 to −25 pcm band.

# 10 Hybrid SVD + Windowed Multipole

## 10.1 Per-nuclide representation byte count

The MIT ENDF/B-VII.1 WMP library is read directly and its on-disk and in-memory payload measured per nuclide, with SVD rank-2 coverage of the smooth remainder. Table 11 reports per-nuclide totals; script `scripts/hybrid_svd_wmp_experiment.py`, raw output `outputs/hybrid_wmp/results.txt`.

**Interpretation, and what the 132.9× does and does not mean.** This table compares two specific data layouts: WMP + rank-2 SVD-smooth against a six-temperature pointwise table, both restricted to four reactions ($\sigma_{\text{el}}, \sigma_{\text{t}}, \sigma_{\text{f}}, \sigma_{\text{cap}}$). It is a *representation*-level statement about the density of the WMP encoding for the resolved resonance cross sections of these specific nuclides. The 132.9× is therefore the density gap between the WMP encoding and a four-channel pointwise table on the matching nuclide set; it is not the reduction a transport engine carrying the full reaction set would see. The engine-scale measurement is in Sec. 10.4.

## 10.2 WMP vs HDF5 accuracy

We evaluate WMP on a 20 000-point log-spaced grid inside each $[E^{\text{WMP}}_{\text{min}}, E^{\text{WMP}}_{\text{max}}]$ at $T = 294$ K and compare to the HDF5 pointwise reference.

Median agreement is $\sim 10^{-4}$, $p_{99}$ is $\sim 10^{-3}$. Maximum absolute errors tail to $10^{-1}$–$10^{0}$ at isolated points for some

**Table 10.** PWR pin cell at +150 K global offset (fuel 900 → 1050 K, mod 600 → 750 K, clad 600 → 750 K), three-way comparison, ten-seed / 50 000-particle / 120-batch statistics. The two-point row is reported as an ablation against the shipped three-point variant.

| Provider | $k_\infty$ | $\sigma_{\text{seed}}$ | Gap vs WMP (pcm) | ns/p | Memory (MB) |
|---|---|---|---|---|---|
| SVD, 2-pt unity Ducru (ablation) | 1.30707 | 0.00037 | −2094 | 15 685 | 152.5 |
| **SVD, 3-pt unity Ducru** | **1.32593** | **0.00059** | **−213** | **27 298** | **152.5** |
| Pointwise + stochastic $T$-pick | 1.32304 | 0.00050 | −502 | 26 451 | 201.3 |
| ACE+WMP (reference) | 1.32806 | 0.00045 | +0 | 27 279 | 199.7 |

| Weighting | rank 3 L2 / peak | rank 5 L2 / peak | rank 5 weights |
|---|---|---|---|
| 3-pt unity Ducru (signed) | 1.05% / 0.994 | **0.84% / 0.990** | $\{-0.12, 0.62, 0.50\}$ |
| 3-pt simplex QP (non-neg) | 5.02% / 1.045 | 5.01% / 1.043 | $\{0.00, 0.56, 0.44\}$ |
| 3-pt kernel-Gram QP (non-neg) | **0.60% / 1.000** | 2.06% / 0.983 | $\{0.00, 0.36, 0.64\}$ |

nuclides (HDF5 grid samples a narrow peak densely while WMP evaluation lies a fraction of the peak width away). These do not carry through to integrated quantities.

## 10.3 End-to-end hybrid engine

We implemented `src/wmp.rs` (CPU WMP reader and Humlicek W4 evaluator) and `src/transport/hybrid_xs.rs` which wraps `SvdXsProvider`. For each nuclide with WMP coverage and each energy in $[E_{\min}^{\text{WMP}}, E_{\max}^{\text{WMP}}]$, the hybrid provider substitutes WMP-evaluated elastic, fission, and capture into the `MicroXs` returned by the SVD provider; inelastic, $(n, 2n)$, $(n, 3n)$ remain on the SVD path. URR overrides are short-circuited inside the WMP window.

**Unit-level validation.** The CPU WMP evaluator matches `openmc.data.WindowedMultipole.__call__` to $\leq 4 \cdot 10^{-6}$ relative error on the three dominant U-238 resonances at $T = 293.6$ K (`rust_prototype/src/bin/wmp_validate.rs`):

| $E$ (eV) | OpenMC $\sigma_{\text{abs}}$ (b) | Rust $\sigma_{\text{abs}}$ (b) | Δ rel |
|---|---|---|---|
| 6.674 | 7108.719 | 7108.695 | $3.3 \cdot 10^{-6}$ |
| 20.9 | 6352.230 | 6352.212 | $2.8 \cdot 10^{-6}$ |
| 36.7 | 5357.922 | 5357.903 | $3.6 \cdot 10^{-6}$ |

**End-to-end $k_\infty$ on PWR pin cell.** Single seed × 100 batches (20 inactive) × 20 000 particles ($1.6 \cdot 10^6$ active histories) on the small-L3 system, with ACE+WMP run as the industry baseline at matching statistics.

The hybrid-vs-WMP gap of −26 pcm sits at $\sim 0.25\,\sigma_{\text{seed}}$, well inside seed noise; we do not read it as a precise correctness claim but it is the right order for what WMP and SVD-of-resonance-data should agree on. WMP represents the resonance cross section by a rational pole fit rather than a piecewise-linear-in-log-log sampling of the same evaluated-nuclear-data curve, so small systematic disagreements in the RRR accumulate at this scale on a nine-nuclide thermal pin cell. The Table-vs-WMP gap of +82 pcm is the cross-representation systematic at the same statistics: pointwise sampling of the same library at the same temperature lands $\sim 0.8\sigma_{\text{seed}}$ from the WMP pole fit. The off-library counterpart of this measurement (Sec. 8.2) tightens $\sigma_{\text{seed}}$ to 59 pcm and resolves the SVD-vs-WMP gap as a real 213 pcm systematic; on-library at this statistics the gap is below noise.

**Throughput.** At small-L3 statistics the hybrid runs at 26 662 ns/p, 0.91× the CPU SVD rank-5 throughput (28 651 ns/p) on the same system; the cache-hit pattern compresses the two providers toward the same wall on the mobile workstation, so the apparent "speedup" is a cache-residency artefact rather than an algorithmic improvement. The large-L3 reference run measures the hybrid at 33 051 ns/p, 2.06× slower than CPU SVD rank-5 (16 052 ns/p); a single WMP evaluation on this hardware is ∼ 67 ns against ∼ 5 ns for SVD, and ∼ 20% of lookups fall in the RRR, so the 2× slowdown is the headline hybrid throughput cost. WMP is a memory-centric representation, and in this engine it costs throughput.

## 10.4 Memory in the actual engine (broad claim)

`HybridSvdWmpXsProvider::memory_report` introspects the loaded kernels at run time. We report four variants for the nine-nuclide PWR set at rank-5: the pointwise-table baseline, pure SVD rank-5 (all reactions compressed), the hybrid in-solver state with the full SVD energy grid, and a smooth-only rebuild in which the elastic, fission, and capture SVD kernels are re-fit on the energy points outside each nuclide's WMP window. The smooth-only rebuild is exposed as `--rebuild-svd-smooth-only` and the numbers below are measured from the loaded kernels at run time.

| | bytes (MB) | vs pointwise |
|---|---|---|
| Pointwise-table baseline (Sec. 6) | 100.6 | 1.0× |
| Pure SVD rank-5 (all reactions) | ∼517.9 | 5.1× larger |
| Hybrid in-solver, full SVD grid (SVD basis + WMP payload) | 519.0 | 5.2× larger |
| Hybrid in-solver, smooth-only rebuild (live) | 487.6 | 4.8× larger |

**What the smooth-only rebuild actually does at run time.** On the standard nine-nuclide PWR set at rank-5 the rebuild drops 31 423.1 KB across MT= 2, MT= 18, and MT= 102 — the three reactions for which WMP supplies an analytical

**Table 11.** Representation-byte accounting for WMP vs four reaction channels of a six-temperature pointwise table. This is a specific narrow comparison; see Sec. 10.4 for the in-engine measurement.

| nuclide | WMP range (eV) | WMP disk | WMP payload | pw full (4 rxn × 6 T) | SVD $k$=2 smooth | ratio |
|---|---|---|---|---|---|---|
| U-234 | 0–1 500 | 34.9 KB | 28.0 KB | 10.2 MB | 12.1 KB | 260.0× |
| U-235 | 0–2 250 | 567.3 KB | 559.1 KB | 53.7 MB | 10.0 KB | 96.6× |
| U-238 | 0–20 000 | 603.0 KB | 594.7 KB | 100.1 MB | 10.7 KB | 169.2× |
| O-16 | $0–2.36 \cdot 10^6$ | 38.5 KB | 30.8 KB | 0.78 MB | 53.3 KB | 9.5× |
| H-1 | $0–2 \cdot 10^7$ | 13.7 KB | 8.2 KB | 155 KB | 0.1 KB | 18.5× |
| Zr-90 | 0–53 500 | 9.8 KB | 4.9 KB | 1.48 MB | 4.7 KB | 156.9× |
| Zr-91 | 0–10 000 | 16.8 KB | 9.5 KB | 3.17 MB | 4.8 KB | 227.7× |
| Zr-92 | 0–71 000 | 21.1 KB | 14.2 KB | 3.97 MB | 4.5 KB | 217.3× |
| Zr-94 | 0–90 000 | 20.7 KB | 14.2 KB | 4.22 MB | 5.2 KB | 222.8× |
| **total** | — | 1.30 MB | 1.23 MB | **177.7 MB** | 0.10 MB | **132.9×** |

**Table 12.** WMP-vs-pointwise relative error, $T = 294$ K, 20 000 log-spaced points per nuclide.

| nuclide | Elastic | | | Absorption | | |
|---|---|---|---|---|---|---|
| | $p_{50}$ | $p_{99}$ | max | $p_{50}$ | $p_{99}$ | max |
| U-234 | $5.0 \cdot 10^{-5}$ | $1.3 \cdot 10^{-3}$ | $1.6 \cdot 10^{-1}$ | $8.6 \cdot 10^{-4}$ | $6.7 \cdot 10^{-2}$ | 0.99 |
| U-235 | $5.4 \cdot 10^{-5}$ | $1.2 \cdot 10^{-3}$ | $8.0 \cdot 10^{-2}$ | $1.9 \cdot 10^{-4}$ | $1.1 \cdot 10^{-3}$ | $9.1 \cdot 10^{-3}$ |
| U-238 | $3.9 \cdot 10^{-5}$ | $1.3 \cdot 10^{-3}$ | $6.8 \cdot 10^{-1}$ | $2.2 \cdot 10^{-4}$ | $8.0 \cdot 10^{-3}$ | 0.88 |
| O-16 | $5.9 \cdot 10^{-5}$ | $3.3 \cdot 10^{-3}$ | $2.4 \cdot 10^{-2}$ | $3.0 \cdot 10^{-4}$ | $1.4 \cdot 10^{-2}$ | $2.0 \cdot 10^{-2}$ |
| H-1 | $4.9 \cdot 10^{-5}$ | $1.3 \cdot 10^{-3}$ | $3.1 \cdot 10^{-3}$ | $2.2 \cdot 10^{-4}$ | $3.7 \cdot 10^{-3}$ | $8.8 \cdot 10^{-3}$ |
| Zr-90 | $5.5 \cdot 10^{-5}$ | $6.7 \cdot 10^{-4}$ | $7.3 \cdot 10^{-2}$ | $2.9 \cdot 10^{-4}$ | $8.0 \cdot 10^{-3}$ | $4.5 \cdot 10^{-1}$ |
| Zr-91 | $2.5 \cdot 10^{-4}$ | $8.8 \cdot 10^{-4}$ | $2.2 \cdot 10^{-2}$ | $1.9 \cdot 10^{-4}$ | $3.5 \cdot 10^{-2}$ | $2.9 \cdot 10^{-1}$ |
| Zr-92 | $2.6 \cdot 10^{-4}$ | $8.9 \cdot 10^{-4}$ | $1.3 \cdot 10^{-1}$ | $3.3 \cdot 10^{-4}$ | $8.0 \cdot 10^{-3}$ | 0.99 |
| Zr-94 | $8.1 \cdot 10^{-5}$ | $7.4 \cdot 10^{-4}$ | $1.1 \cdot 10^{-1}$ | $2.9 \cdot 10^{-4}$ | $8.6 \cdot 10^{-3}$ | 0.90 |

representation inside the resonance window, so the SVD basis no longer needs energy points there. Discrete-level kernels (MT= 51–91) and other channels are untouched. The full sequence is reported by the engine on every hybrid run:

```
Before smooth-only rebuild:
  full SVD basis = 517 870.5 KB
  WMP payload    =   1 153.8 KB
  TOTAL          = 519 024.3 KB

After smooth-only rebuild (live):
  smooth SVD     = 486 447.4 KB
  WMP payload    =   1 153.8 KB
  TOTAL          = 487 601.2 KB
  dropped        =  31 423.1 KB
                   (el/fis/cap)
  reduction      = 1.06x
```

**Where the bytes actually live.** The dominant contribution in every SVD-based variant is the discrete inelastic-level cascade: 41 MT= 51–91 levels per uranium nuclide, 10–13 per zircaloy isotope, each carried as its own rank-5 SVD kernel. On the GPU that cascade alone measures 251.3 MB — by itself 2.5× the entire pointwise-table engine. The smooth-channel SVD bases (elastic, fission, capture, total inelastic, $(n, 2n)$, $(n, 3n)$) together measure 64.6 MB; the WMP payload is 1.23 MB. So switching the resonance representation to WMP costs essentially zero memory, and the smooth-only rebuild's 1.06× reduction captures only the elastic/fission/capture energy points masked by WMP windows — everything else stays. The engine-scale penalty against the pointwise table is not caused by WMP and is not relieved by it; it is caused by storing the discrete-level cascade as per-level rank-5 bases rather than per-level scalar arrays.

**Where the representation-byte ratio still applies.** Equation 7 from Sec. 2.5 says the SVD-vs-pointwise memory ratio is $k/T$ for a single reaction. The engine-scale penalty reported above is not a counter-example: our pointwise-table baseline is built at a single effective temperature per material region (one temperature per nuclide-in-material). In a multi-temperature library ($T = 6$ in the HDF5 files shipped for depletion feedback), the pointwise baseline would grow to $\sim 600$ MB on the same nuclide set while pure SVD and hybrid stay at $\sim 519$ MB; the ordering would then invert. The engine-scale loss reported above is therefore specific to $T = 1$, not a universal property.

The hybrid engine is 4.8–5.2× *larger* than the pointwise-table baseline, because SVD kernels for the 41 discrete levels per U nuclide and 10–13 per Zr isotope dominate memory, and those reactions are not covered by the public WMP library. The smooth-only rebuild (1.06× vs the full grid) is essentially a rounding error.

The 132.9× ratio in Table 11 is a correct and reproducible *representation*-byte comparison — WMP is denser than a

**Table 13.** End-to-end SVD+WMP hybrid on PWR pin cell, single seed × 100 batches × 20 000 particles, small-L3 (i7-12800H). ACE+WMP is the industry-baseline reference; SVD and Table rows from Table 8 are repeated here for direct comparison.

| mode | rank | $k_\infty$ | $\sigma_{\text{seed}}$ | $\Delta$ vs ACE+WMP (pcm) | ns/p |
|---|---|---|---|---|---|
| ACE+WMP (industry baseline) | — | 1.32821 | 0.00098 | +0 (ref) | 26 317 |
| CPU SVD | 5 | 1.32817 | 0.00100 | −4 | 29 273 |
| **Hybrid SVD+WMP** | **5** | **1.32795** | **0.00103** | **−26** | **26 662** |
| CPU table | — | 1.32903 | 0.00093 | +82 | 26 035 |

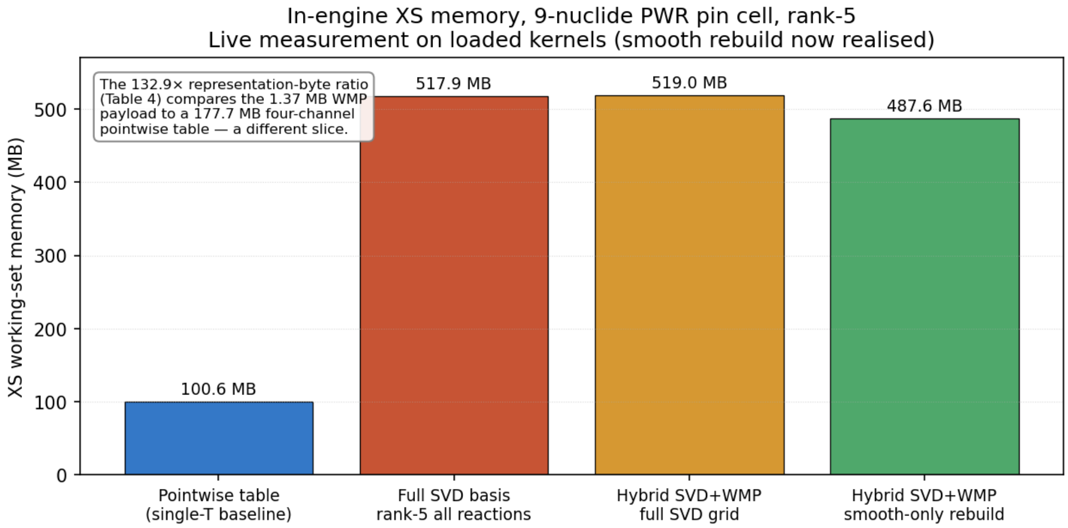


**Figure 9.** In-engine XS working-set memory for the nine-nuclide PWR pin cell, rank-5, measured live on the loaded kernels. The smooth-only rebuild (rightmost bar) is now realised in the engine via `--rebuild-svd-smooth-only`: it drops 31.4 MB across MT= 2, MT= 18, and MT= 102 for a 1.06× reduction. The hybrid variants both carry more memory than the pointwise-table baseline because the SVD kernels for the discrete inelastic levels, not covered by WMP, dominate the working set. The 132.9× representation-byte ratio of Table 11 is a different comparison (1.37 MB WMP vs 177.7 MB four-channel pointwise).

four-channel pointwise table at matched energy range. But in a transport engine that carries the full reaction set, WMP does not shrink the working set unless the rest of the reaction set is also compressed. A version of this hybrid that achieved representation-scale memory reduction would need: (i) WMP coverage for the discrete inelastic cascade (MT= 51–91); or (ii) a different representation for the cascade (e.g. ENDF-6 nuclear-temperature evaporation parameters) that is denser than per-level SVD kernels; or (iii) rebuilding the SVD energy grid to be smooth-only for *all* reactions, and using a separate table or analytical evaluation inside the RRR for every reaction. The smooth-only rebuild implemented here is item (iii) restricted to MT= 2, MT= 18, MT= 102; extending it to the rest of the reaction set is engineering work.

**Hybrid status.** (a) WMP is the correct analytical representation for the RRR and the engine ships a pure-Rust evaluator (matches OpenMC Python at $\sim 10^{-6}$ relative error, CUDA port matches to machine precision); (b) end-to-end $k_\infty$ at single-seed statistics agrees with ACE+WMP within −26 pcm ($\ll 1\sigma_{\text{seed}}$); (c) for a full-core engine where the cascade reactions are stored more densely than per-level SVD (e.g. nuclear-temperature evaporation), the RRR saving becomes proportionally more valuable. The hybrid is shipped as a correctness deliverable, not as a memory or throughput win on the present engine.

## 11 GPU WMP evaluator

WMP evaluation is compute-dense (Faddeeva + broadened curvefit + pole sum) with a small working set (few KB of pole data per nuclide), which matches GPU arithmetic pipelines well. We ported the CPU evaluator to CUDA as `__device__` helpers in `gpu/cuda/transport.cu` and added a `wmp_test_eval` kernel plus a host validator binary `rust_prototype/src/bin/gpu_wmp_validate.rs`.

**Bit-exactness.** CPU-vs-GPU over twelve U-238 test energies at $T = 293.6$ K: maximum relative discrepancy $5.3 \cdot 10^{-14}$ on absorption and $2.0 \cdot 10^{-13}$ on fission — double precision round-off only.

**Throughput on RTX** 3080. On a $10^6$-energy log-spaced grid inside U-238's RRR:

| Implementation | ns/lookup | Relative |
|---|---|---|
| CPU single-thread (Ryzen 9800X3D) | 66.8 | 1.00× |
| GPU (RTX 3080, saturated) | 9.5 | 0.14× |

7.0× per-lookup GPU-vs-CPU.

**Standalone GPU evaluator only.** The validation above is for the GPU evaluator on its own, exercised by a host driver that broadcasts a fixed energy grid to all threads and reads back the cross-section vector. Wiring the evaluator into the event-based `transport_persistent` kernel for end-to-end GPU hybrid transport requires a bitmap `has_wmp[i]` plus per-nuclide $E_{\min}^{\mathrm{WMP}}$, $E_{\max}^{\mathrm{WMP}}$ checks inline, the recursive Humlicek conjugate fold unrolled to a flag-plus-sign-flip form (recursive `__device__` calls spill to local memory under the register pressure of `__launch_bounds__(256, 2)` and trigger CUDA_ERROR_ILLEGAL_ADDRESS on deep stacks), and a recomputed total cross section from partials with URR sampling suppressed inside the WMP window. We have not completed the in-kernel integration and report no in-engine GPU hybrid $k_\infty$ or throughput.

# 12 Adjoint-flux compression for weight-window storage

The same SVD machinery used for cross-section compression (Section 2) applies — with a one-line code change — to the random-ray adjoint flux $\psi^*(\mathbf{r}, g)$ used by FW-CADIS weight-window generation [13]. The adjoint solver returns $\psi^*$ as a row-major $[n_{\mathrm{FSR}} \times n_g]$ matrix; for Cartesian voxel meshes this reshapes naturally as $[n_x\, n_y \times n_z]$, separating the streaming axis from the transverse plane. Slab and duct geometries put almost all of $\psi^*$ in a single dominant singular triplet of this reshape, leaving the rest of the spectrum to capture statistical noise from the MC adjoint solve.

## 12.1 Adaptive picker

Rather than picking a fixed rank, the implementation uses an adaptive picker: given a Frobenius reconstruction tolerance $\varepsilon$, return the lowest-rank SVD whose linear-space reconstruction satisfies $\|\hat{\psi}^* - \psi^*\|_F / \|\psi^*\|_F \le \varepsilon$ and whose factored byte count $(m+n+1)\cdot k \cdot 8$ is strictly below the dense baseline $m \cdot n \cdot 8$. If no rank in $1 \ldots k_{\max}$ qualifies, the picker falls back to the dense matrix. Both linear-space and $\log_{10}$-space SVD are tried; the smaller representation that meets $\varepsilon$ wins. The dense fallback is byte-identical to the existing CADIS map JSON the photon-side weight-window pipeline already consumes, so the switch is invisible to downstream code.

## 12.2 Mesh-size sweep

A meaningful sweep needs the voxels to be coarser than the underlying physics: subdividing finer than ~mfp/8 introduces no signal, only noise from the random-ray sampling. For water at 1 MeV (mfp ≈ 14 cm) the floor is ≈ 1.77 cm, so the sweep is restricted to meshes whose smallest voxel edge is ≥ mfp/8. The sweep binary refuses sub-mfp meshes at the source so that pollution can't sneak into the analysis.

Figure 10 runs 23 such meshes through the picker at two Frobenius tolerances. Voxel counts span 25 to $1.25 \times 10^5$ — a 3.7-decade range bounded above by the physics-resolution floor. Numbers are `[micro]`-scope — they characterise the compression kernel on real solver output, not end-to-end shielding tally throughput. Random-ray statistics are fixed at 200 × 8000 rays for every mesh.

| mesh | voxels | dense | $\varepsilon$ | picked | ratio |
|---|---|---|---|---|---|
| *$\varepsilon$ = 2.5% — selected sweep points* | | | | | |
| 1×1×25 | 25 | 200 B | 2.5 % | $k$=2 | 1.14× |
| 5×5×25 | 625 | 5.0 KB | 2.5 % | $k$=2 | 6.13× |
| 10×10×25 | 2 500 | 20 KB | 2.5 % | $k$=3 | 6.61× |
| 20×20×50 | 20 000 | 156 KB | 2.5 % | $k$=4 | **11.09**× |
| 30×30×25 | 22 500 | 176 KB | 2.5 % | $k$=3 | 8.10× |
| 35×35×25 | 30 625 | 239 KB | 2.5 % | $k$=3 | 8.16× |
| 30×30×50 | 45 000 | 351 KB | 2.5 % | $k$=6 | 7.89× |
| 50×50×25 | 62 500 | 488 KB | 2.5 % | $k$=5 | 4.95× |
| 40×40×50 | 80 000 | 625 KB | 2.5 % | $k$=8 | 6.06× |
| 50×50×50 | 125 000 | 977 KB | 2.5 % | $k$=8 | 6.13× |
| *$\varepsilon$ = 1.0% — selected sweep points* | | | | | |
| 5×5×25 | 625 | 5.0 KB | 1.0 % | $k$=3 | 4.09× |
| 20×20×50 | 20 000 | 156 KB | 1.0 % | $k$=10 | 4.43× |
| 30×30×25 | 22 500 | 176 KB | 1.0 % | $k$=7 | 3.47× |
| 40×40×50 | 80 000 | 625 KB | 1.0 % | $k$=14 | 3.46× |
| 50×50×50 | 125 000 | 977 KB | 1.0 % | $k$=15 | 3.27× |
| 50×50×25 | 62 500 | 488 KB | 1.0 % | dense | 1.00× |

**Table 14.** Picker decisions on the 23-mesh physics-respecting sweep (smallest voxel edge ≥ mfp/8 ≈ 1.77 cm), slab benchmark, 200 × 8000 mortal rays. Selected rows; full data in `outputs/adjoint_sweep_valid_tol{1,2.5}pct.csv`. At $\varepsilon$ = 2.5% every mesh in the meaningful range compresses, in a 5–11× band. Tightening to $\varepsilon$ = 1% shrinks the picked rank's headroom and flips a few of the largest meshes to the dense fallback.

The compression ratio rises with mesh size as the picker fills out the rank-1 dominant streaming mode and stabilises in a ~ 5–11× band once the mesh is large enough to saturate that mode. At $\varepsilon$ = 2.5% every mesh in the physically meaningful range compresses; the picker chooses ranks that grow gradually with voxel count (typically $k$=2–8). At $\varepsilon$ = 1% the picker still wins on most meshes but falls back to dense at 50×50×25 — the random-ray noise floor at 200 × 8000 rays catches up with the tolerance there. In a deployment scenario the picker simply picks the rank that satisfies the user's accuracy budget or returns dense — there is no manual tuning and no risk of a silent accuracy regression. For the existing `outputs/cadis_water_*.json` files ($1 \times 1 \times n_z$ mesh, $n_z \le 25$) the picker chooses dense or a small rank-2 SVD with ~ 1.1× savings; the win arrives only when 3D voxelisation is enabled, as the literature predicted, and grows to the 5–11× band across the meshes where $\psi^*$ compression actually matters for storage budgets.

Sub-mfp meshes were tried during exploration but produce no meaningful result — the geometric subdivision is finer than the random-ray sampling can resolve, so the picker is forced to fit noise. The sweep binary refuses sub-mfp meshes at the source (asserting that the smallest voxel edge be ≥ mfp/8) so that the analysis cannot be polluted by configurations where the physics carries no signal.

## 12.3 Approaches that did not pan out

Two alternative compression strategies were implemented and benchmarked against the SVD picker, but neither displaced it.

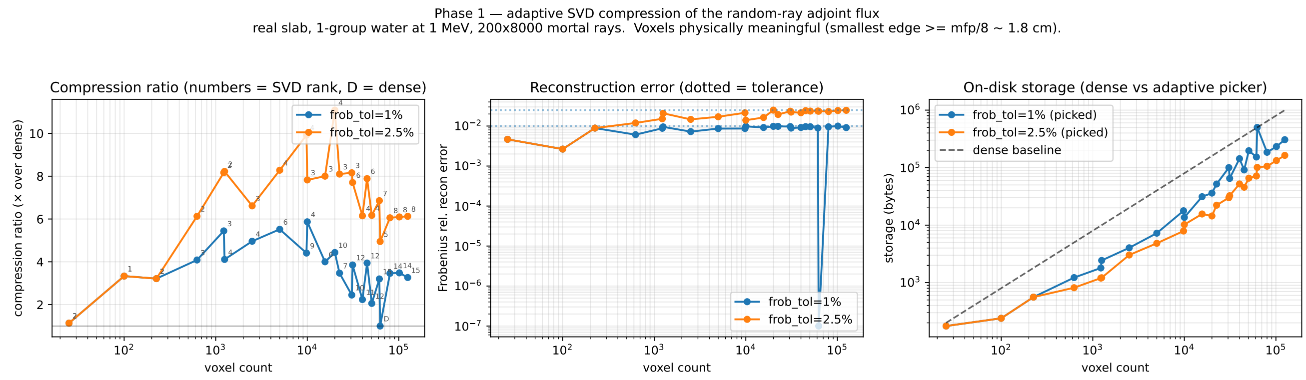


**Figure 10.** Adaptive SVD compression of the random-ray adjoint flux on the slab benchmark, water at 1 MeV, 1-group, mortal rays ($200 \times 8000$). 23 mesh configurations restricted to physically meaningful voxel resolutions (smallest edge $\geq$ mfp/8 $\approx$ 1.77 cm); voxel counts span 25 to $1.25 \times 10^5$. Annotations are SVD rank chosen (`D` = dense fallback). *Left*: compression ratio at $\varepsilon = 2.5\%$ rises to ~8–11× between $10^3$ and $10^5$ voxels and stays in that band; the peak is 11.09× at $20 \times 20 \times 50 = 20\,000$ voxels. *Middle*: linear reconstruction error stays under the requested tolerance band on every retained SVD pick; dense fallbacks are off-scale (exact, plotted at the $\varepsilon$ floor). *Right*: storage in bytes — picked vs dense.

**Tensor-train decomposition.** The 3-tensor $\Sigma(E, T, R)$ for a single nuclide can in principle be compressed jointly via TT-SVD [14] rather than per-reaction 2D SVD. On a synthetic $5000 \times 9 \times 52$ tensor with planted CP-rank 5, TT at rank $(5, 5)$ used 51× less memory than the per-reaction baseline and matched it on accuracy. On real U-235 data with $3000 \times 6 \times 6$ (six reactions), TT at rank 5 was 6× smaller but $\approx 10^3$× less accurate than the production per-reaction 2D SVD at the same nominal rank ($4.8 \times 10^{-3}$ vs $4.2 \times 10^{-6}$ Frobenius rel error). TT loses one degree of freedom in the joint structure, and on the small reaction counts and high accuracy budgets relevant to MC $k_{\text{eff}}$ that loss is not tradeable for the memory win. This is consistent with the prior cross-isotope basis-sharing result (subspace angle $> 85°$ at $k = 2$) but rules out a stronger method as well.

**Discrete cosine transform of $\sigma(E, T)$.** A 2D DCT-II of $\log \sigma(E, T)$ truncated to the top-$K$ coefficients was benchmarked head-to-head against SVD on U-235 fission MT 18 across five byte budgets from 8 KB to 128 KB. SVD won at every budget tested. Off-library reconstruction at $T_q$ = 750 K (between the 600 K and 900 K library columns) gave SVD $4.7 \times 10^{-3}$ relative error at 16 KB while DCT achieved $2.1 \times 10^{-2}$; at 128 KB SVD becomes machine-exact while DCT remains at $1.7 \times 10^{-3}$. The off-library memory edge of SVD reported in Section 7 is therefore not eliminated by a clean transform-and-truncate alternative. A heat-equation formulation [1] of DCT-DB might do better, but at WMP-scale implementation effort this was not in scope; the simple alternative is dominated.

The full benchmark CSVs and numbers for both negative results live at `outputs/phase2_tt_decompose.txt` and `outputs/phase3_dct_db.txt`.

## 13 Threats to validity

**Scope and nature of benchmarks.** Two systems: Godiva (HEU-MET-FAST-001, three nuclides, fast spectrum, *physical* ICSBEP benchmark with measured $k_{\text{eff}}^{\text{exp}} = 1.0000 \pm$ 100 pcm $1\sigma$) and a nine-nuclide thermal PWR pin cell (no physical benchmark; OpenMC 0.15.3 is used as an independent code-to-code reference). Full-core LWR analyses carry $\gtrsim$ 200 nuclides; the hybrid memory picture scales with the reaction-channel mix and could look very different there. On Godiva $\Delta_{\text{exp}}$ against ICSBEP is the primary validation metric; $\Delta_{\text{OMC}}$ against OpenMC is reported as an independent cross-code check; the two codes straddle experiment from opposite sides at this statistics, and the cross-code spread (178 pcm) is larger than either code's offset from the measurement. On PWR, with no physical benchmark available, the OpenMC $|\Delta| \leq 51$ pcm agreement is the strongest claim we can make and should be read as a code-to-code cross-check, not as absolute validation.

**S($\alpha$,$\beta$) coverage.** Only H in $H_2O$ is exercised. $D_2O$, graphite, ZrH, and other thermal libraries are not validated.

**Sampling convention.** The angular and fission outgoing-energy distributions use stochastic-bin selection, matching OpenMC. A fully independent reference (MCNP, Serpent) would reveal whether OpenMC itself is biased; this is not investigated here.

**Charged-particle absorption.** Channels MT= 103, 107, first-chance fission (MT= 19–21) are absorbed into the capture macroscopic. $k_\infty$ is correct under this convention; MT-level reaction-rate tallies are not.

**Statistical confidence.** All comparisons use SEM $= \sigma_{\text{seed}}/\sqrt{N_{\text{seeds}}}$, not $\sigma_{\text{seed}}$ itself. Large-L3 sweeps use $N_{\text{seeds}}$ = 10; hybrid sweep uses five.

**Hybrid memory.** The 132.9× representation-byte ratio is not an in-engine reduction. In-engine reduction is 1.06× (Sec. 10.4). The 132.9× density ratio is reported for what it is; the 1.06× engine-level reduction is the number an engine builder would care about.

**GPU SVD: per-warp cache, falsified and removed.** Earlier drafts identified a per-warp $(n, E_{\mathrm{idx}})$ shared-memory cache as the engineering item that would close the GPU SVD vs GPU pointwise gap on PWR. We implemented it, measured it on RTX A1000 at five seeds × 60 batches × 50 000 particles, and obtained 1.003× throughput with bit-parity within SEM — the cache was correct but inert because energy-bin sort granularity (256 bins across 17 decades) does not produce warp-level $(n, E_{\mathrm{idx}})$ uniformity often enough to activate the slot. The hypothesis is falsified at this geometry. The actual fix turned out to live upstream: *synthesise MT= 4 at HDF5 load and tabulate the per-level CDF on a log-decimated grid*, which together remove the 13–41-deep level walk entirely (Sec. 9). The cache code has been removed; the falsification is documented here.

**GPU hybrid.** Only the WMP evaluator itself is validated on GPU; the transport-loop integration is not done. No in-engine GPU hybrid $k_\infty$ or throughput is reported.

# 14 Related work

Adaptive resampling machine-learned models [3, 4] retain 10–50% of the pointwise data with ≤ 97 pcm criticality error. Our approach is orthogonal (replace the representation, not subsample the array). Windowed Multipole for Doppler broadening is due to [5, 6]; hybrid multipole + windowed networks in RMC code [7] pair multipole with a second analytical method; our hybrid pairs multipole with SVD. Event-based GPU transport follows the Tramm architecture [8] and shares the memory-bottleneck diagnosis of earlier XSBench studies [9]. Cross-isotope basis sharing was investigated and found not viable beyond the first singular vector (subspace angle $> 85°$ at $k = 2$ between $^{235}$U, $^{238}$U, $^{239}$Pu), a finding that echoes KV-cache compression experience in a different domain [15].

# 15 Conclusion

Truncated SVD of $\log_{10} \sigma(E, T)$ is a viable drop-in for pointwise tables in continuous-energy Monte Carlo neutron transport, with the following constraints satisfied: rank 5 on the smooth channels, rank 1 on the MT= 51–91 discrete-level cascade, three-point unity-normalized Ducru weights for off-library temperature interpolation, and channel-consistent stochastic $T$-picking. Validation against ICSBEP HEU-MET-FAST-001 lands at +79 ± 38 pcm, inside the experimental ±100 pcm uncertainty band; on a 9-nuclide PWR pin cell, agreement with the engine's own pointwise table is 5–47 pcm and with OpenMC 0.15.3 on the same library is $|\Delta| \leq 51$ pcm at ten-seed statistics.

## 15.1 When SVD pays

**Off-library temperatures.** This is the regime where SVD's measurement story is unambiguously positive. At a uniform +150 K offset on PWR, SVD with three-point unity Ducru is 22% smaller than ACE+WMP in memory (156 vs 206 MB), matches its CPU throughput within 0.07%, and trails its $k_\infty$ by 213 pcm ($\sim 3.6\sigma_{\mathrm{seed}}$, Sec. 8.2). The pointwise table doubles its memory off-library because stochastic $T$-picking requires both bracketing library columns to be resident; SVD reconstructs the off-library $\sigma$ from a single basis with three weights and does not pay this cost.

**High rank-1 reaction-channel mix.** 43 of 47 U-235 reaction channels are rank-1 to machine precision (Sec. 4.1); the weakly-$T$-dependent MT= 51–91 discrete levels are each rank-1 to machine precision across all seven ENDF/B-VII.1 library columns. Where the reaction-channel mix matches this profile, the rank budget is small and the in-engine memory penalty for SVD is correspondingly small.

**Multi-temperature libraries.** With $T = 6$ library columns (the standard depletion-feedback multi-temp HDF5 set), the pointwise baseline grows linearly in $T$ to ~600 MB on the 9-nuclide set while pure SVD and hybrid stay at ~519 MB; the on-library memory ordering inverts in SVD's favour. The measurements in this paper use $T = 1$, so this is a projection, not a measured win.

## 15.2 Where SVD does not pay

**On-library, $T = 1$, full-physics deployment.** At every rank measured (1, 2, 3, 5, 7), SVD in-engine memory is monotonically larger than the pointwise table at on-library temperatures (Sec. 7). The basis stores $r$ values per energy point; the table stores one. CPU throughput is favourable (1.37×–1.90× vs the in-engine pointwise table on PWR), but on memory there is no on-library rank that wins.

**GPU.** GPU SVD on PWR initially measured 1.30× slower than GPU pointwise on the same geometry. The cause was the level- sum walk for nuclides whose ENDF/B-VII.1 evaluation omits MT= 4 (Zr-90/91/92/94, U-238): the macroscopic inelastic XS had to be synthesised at lookup time by summing 13–41 discrete-level SVD kernels per step. A per-warp $(n, E_{\mathrm{idx}})$ shared-memory cache, our first hypothesis for the fix, was implemented and falsified (1.003× at five seeds; energy-bin sort coherence too low; preserved as a cautionary result in §13). The actual fix lives at HDF5 load: synthesise MT= 4 from the discrete-level sum and tabulate the per-level CDF on a log-decimated grid. Both together collapse the kernel hot path to a single rank-5 `svd_reconstruct` plus a binary CDF search. On RTX A1000 the synth+CDF row at three seeds × 60 batches × 50 000 particles runs 1.0099× *faster* than GPU pointwise on the same geometry (19 473 vs 19 664 ns/p), with $k_\infty$ agreeing to +16 pcm (well inside SEM).

**Hybrid SVD+WMP throughput.** The hybrid is 2.06× slower than CPU SVD because the Faddeeva evaluation inside each WMP window is ~ 15× more expensive than the SVD FMA. The hybrid is shipped as a correctness deliverable, not as a throughput improvement; it is the right path only when WMP-grade resonance accuracy is required and the 2× slowdown is acceptable.

### 15.3 Caveats and limits

**Engine-scale memory ≠ representation-byte ratio.** The 132.9× representation-byte ratio of WMP+smooth-SVD against a hypothetical four-channel multi-temperature pointwise table is correct per-nuclide-per-channel (Table 11), but the in-engine working set on PWR is 4.8–5.2× *larger* than the pointwise baseline at $T = 1$, because the discrete-level SVD kernels (which WMP does not cover) dominate the engine memory. The smooth-only-rebuild reduces this by 1.06× (519.0 → 487.6 MB). Quoting a single compression ratio without scope conflates these two metrics; both are reported here so the comparison is unambiguous.

**Rank-1 on PWR collapses.** Rank-1 SVD reconstructs U-238 resonance self-shielding to a single basis vector and produces $k_\infty = 0.871$ on the on-library PWR pin cell (~ 46 000 pcm low). Rank-2 overshoots (1.409, ~ 8 000 pcm high). Rank-3 is the lowest deployable rank on resonance-dominated thermal systems; rank-5 is the production floor.

**Library-bias not investigated.** Both the engine and OpenMC sit inside the ICSBEP ±100 pcm uncertainty band on Godiva, from opposite sides; the +178 pcm cross-code spread between them is larger than either's offset from experiment. A fully independent reference (MCNP, Serpent) would localise the spread to one code or to the underlying ENDF/B-VII.1 evaluation. This is not investigated here.

**S$(\alpha,\beta)$ coverage limited to H/H$_2$O.** D$_2$O, graphite, ZrH and other thermal libraries are not validated against an independent reference at this stage.

**Charged-particle channels absorbed into capture.** Channels MT= 103, 107, and first-chance fission (MT= 19–21) are folded into the capture macroscopic. $k_\infty$ is correct under this convention; MT-level reaction-rate tallies are not.

**Single-seed hybrid statistics.** The end-to-end Hybrid SVD+WMP $k_\infty = 1.32795 \pm 0.00103$ (−26 pcm vs ACE+WMP) is single-seed only at the matched 150×50 000 statistics; the ten-seed redo is not in this paper.

**CP/PARAFAC of $\sigma(E,T,\ell)$ — measured, deferred.** The per-level rank-1 SVDs we currently store ignore cross-$\ell$ correlation. We probed CP/PARAFAC — a joint rank-$R$ decomposition $\sigma_\ell(E,T) \approx \sum_{r=1}^{R} a_r(E)\, b_r(T)\, c_r(\ell)$ — via greedy rank-1 deflation (`src/cp_decompose.rs`) and report results below at 200 log-spaced energy points across the six ENDF/B-VII.1 library temperatures (`outputs/cp_analysis.txt`).

| nuclide | $N_\ell$ | rank-3 rel. L2 | rank-5 rel. L2 | rank-8 rel. L2 | mem ratio at rank-8 |
|---|---|---|---|---|---|
| Zr-90 | 10 | 7.0% | 1.4% | 0.26% | 0.84× |
| Zr-91 | 10 | 3.6% | 0.78% | 0.046% | 0.84× |
| Zr-92 | 10 | 8.6% | 2.9% | 0.32% | 0.84× |
| Zr-94 | 13 | 9.8% | 3.5% | 0.82% | 0.66× |
| U-238 | 41 | 6.0% | 1.9% | 0.45% | **0.23×** |

The pattern is the predictable one: CP wins as $N_\ell$ grows because the cost of the $\ell$-mode factor ($N_\ell$ doubles per rank) is amortised against the constant overhead of the $E$-mode and $T$-mode factors. For U-238 at rank-8, CP reaches sub-pcm RMSE while storing 4.3× less than the 41 per-level rank-1 SVDs. For Zr clad with 10–13 levels, CP only barely breaks even on memory at the precision we care about.

We do not deploy CP because synthesis + CDF already crossed parity with GPU pointwise at the per-particle cost, and CP loses that comparison: a rank-$R$ CP reconstruction at one $(E_{\text{idx}}, T)$ costs $R(N_\ell + 2)$ FMAs to materialise all levels at once ($R = 5, N_\ell = 41 \rightarrow 215$ FMAs) versus the synthesis-plus-CDF hot path's one rank-5 `svd_reconstruct` plus one binary CDF search (~ 18 FMA-equivalents). CP becomes the right pick for full-actinide libraries with many high-level nuclides (Pu isotopes, minor actinides), and for analyses where all 13–41 level cross sections are queried in batch (sensitivity tallies, depletion chains). Both are out of scope here.

### 15.4 Bottom line

SVD compression of ENDF/B-VII.1 cross sections is the right choice for off-library, multi-temperature, or rank-1-channel-heavy operating regimes on CPU; the pointwise table remains the right choice for on-library, single-temperature, GPU-deployed, or memory-constrained configurations. The signed three-point unity Ducru scheme is the production default at rank 5; the kernel-Gram QP wins at low rank but loses at the production rank. Three structural properties of ENDF/B-VII.1 — the rank-1 majority of U-235 reaction channels, the weakly-$T$-dependent discrete-level cascade, and the rank deficiency of $\log \sigma(E,T)$ at fixed $E$ — are real and exploitable; the raw Ducru weights are not a partition of unity and must be normalized before being applied to a $\log \sigma$ SVD.

## Data availability

The transport engine (pure Rust), the CUDA event-based solver, the CUDA WMP evaluator, all input decks, per-seed benchmark CSV files, the XS-audit scripts, the hybrid experiment Python script, and the scripts that generate the figures in this paper are released with the preprint. ENDF/B-VII.1 neutron and thermal data are available from the OpenMC data archive; the ENDF/B-VII.1 Windowed Multipole library is the public MIT release.

## Declaration of AI assistance

This work was written with assistance from large language models for code and prose drafting; all physics, numerical experiments, and results were reviewed and validated by the human author.

## References

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